Biomath Booklet

Chapter 3

Contents

		Lecture Problems	Textbook Problems	Page
3.1	Derivative Formulas for Powers and Polynomials	10	63	**3**
3.2	Exponential and Logarithmic Functions	6	50	**24**
3.3	The Chain Rule	7	53	**38**
3.4	The Product and Quotient Rules	5	47	**52**
3.5	Derivatives of Periodic Functions	4	30	**66**

Each section is split into topics. Every topic has its key formulas, then **Lecture Problems**, then **Textbook Problems**. Worked answers are at the end of each section.

3.1 Derivative Formulas for Powers and Polynomials

In this section

1. Constant, Linear and Constant Multiple 3 lecture · 6 textbook
2. The Power Rule 2 lecture · 12 textbook
3. Derivatives of Polynomials 1 lecture · 21 textbook
4. Equation of the Tangent Line 1 lecture · 3 textbook
5. Rate of Change and Applications 2 lecture · 19 textbook
6. The Second Derivative 1 lecture · 2 textbook

3.1.1 Constant, Linear and Constant Multiple

Constant	If $f(x) = k$, then $f'(x) = 0$
Linear	If $f(x) = b + mx$, then $f'(x) = m$ (slope)
Constant multiple	$\frac{d}{dx}\big(cf(x)\big) = cf'(x)$

Lecture Problems

Ex 1 Find $\frac{d}{dx}(5)$.

Ex 2 Find $\frac{d}{dx}(2x + 3)$.

Ex 3 Let $f(x) = 5 + 7x$. Find $\frac{d}{dx}\big(2f(x)\big)$.

Textbook Problems

Find the derivative. Assume a, b, c, k are constants.

1 $y = 5$

2 $y = 3x$

8 $y = 5x + 13$

29 $f(x) = kx^2$

35 $w = 3ab^2q$

36 $h(x) = \dfrac{ax + b}{c}$

3.1.2 The Power Rule

Power rule $\dfrac{d}{dx}\left(x^n\right) = nx^{n-1}$

Lecture Problems

Ex 4 Use the power rule to find:

i. $\dfrac{d}{dx}\left(x^3\right)$

ii. $\dfrac{d}{dx}\left(x^{-5}\right)$

iii. $\dfrac{d}{dx}\left(x^{3/2}\right)$

Ex 5 Find the derivative of

i. $f(x) = x^8$

ii. $h(t) = t^7$

iii. $g(x) = \sqrt{x}$

iv. $w(t) = \dfrac{1}{t^3}$

v. $R(x) = 2x^{4.5}$

Textbook Problems

Find the derivative. Assume a, b, c, k are constants.

3 $y = x^{12}$

4 $y = x^{-12}$

5 $y = x^{4/3}$

6 $y = 8t^3$

9 $f(x) = \dfrac{1}{x^4}$

17 $g(t) = \dfrac{1}{t^5}$

18 $f(z) = -\dfrac{1}{z^{6.1}}$

19 $y = \dfrac{1}{r^{7/2}}$

20 $y = \sqrt{x}$

21 $h(\theta) = \dfrac{1}{\sqrt[3]{\theta}}$

22 $f(x) = \sqrt{\dfrac{1}{x^3}}$

62 Show that for any power function $f(x) = x^n$, we have $f'(1) = n$.

3.1.3 Derivatives of Polynomials

Sum / difference	$\dfrac{d}{dx}\big(f(x) \pm g(x)\big) = f'(x) \pm g'(x)$
Power rule	$\dfrac{d}{dx}\left(x^n\right) = nx^{n-1}$

Lecture Problems

Ex 6 Differentiate

i. $r(p) = p^5 + p^3$

ii. $f(x) = 5x^2 - 7x^3$

iii. $g(t) = \dfrac{t^2}{4} + 3$

Textbook Problems

Find the derivative. Assume a, b, c, k are constants.

7 $y = 3t^4 - 2t^2$

10 $f(q) = q^3 + 10$

11 $y = x^2 + 5x + 9$

12 $y = 6x^3 + 4x^2 - 2x$

13 $y = 3x^2 + 7x - 9$

14 $y = 8t^3 - 4t^2 + 12t - 3$

15 $y = 4.2q^2 - 0.5q + 11.27$

16 $y = -3x^4 - 4x^3 - 6x + 2$

23 $y = 3t^5 - 5\sqrt{t} + \dfrac{7}{t}$

24 $y = z^2 + \dfrac{1}{2z}$

25 $y = 3t^2 + \dfrac{12}{\sqrt{t}} - \dfrac{1}{t^2}$

26 $h(t) = \dfrac{3}{t} + \dfrac{4}{t^2}$

27 $y = \sqrt{x}\,(x + 1)$

28 $h(\theta) = \theta\left(\theta^{-1/2} - \theta^{-2}\right)$

30 $y = ax^2 + bx + c$

31 $Q = aP^2 + bP^3$

32 $v = at^2 + \dfrac{b}{t^2}$

33 $P = a + b\sqrt{t}$

34 $V = \frac{4}{3}\pi r^2 b$

54 Suppose W is proportional to r^3. The derivative dW/dr is proportional to what power of r?

38 Let $f(x) = x^3 - 4x^2 + 7x - 11$. Find $f'(0)$, $f'(2)$, $f'(-1)$.

3.1.4 Equation of the Tangent Line

Tangent line $y - y_1 = m(x - x_1)$, where $m = f'(x_1)$

Lecture Problems

Ex 7 Find an equation for the tangent line at $x = 1$ to the graph of $y = x^3 + 2x^2 - 5x + 7$. Sketch the graph of the curve and its tangent line on the same axes.

Textbook Problems

49 Find the equation of the line tangent to the graph of f at $(1, 1)$, where f is given by $f(x) = 2x^3 - 2x^2 + 1$.

50

(a) Find the equation of the tangent line to $f(x) = x^3$ at the point where $x = 2$.

(b) Graph the tangent line and the function on the same axes. If the tangent line is used to estimate values of the function, will the estimates be overestimates or underestimates?

51 Find the equation of the line tangent to the graph of $f(t) = 6t - t^2$ at $t = 4$. Sketch the graph of $f(t)$ and the tangent line on the same axes.

3.1.5 Rate of Change and Applications

Rate of change	Rate of change at $t = a$: $f'(a)$
Relative rate	Relative rate of change at $t = a$: $\dfrac{f'(a)}{f(a)} \times 100\%$

Lecture Problems

Ex 8 Find the rate of change of a population of size $P(t) = t^3 + 4t + 1$ at time $t = 2$.

Ex 9 Experiments indicate that when a flea jumps, its height (in meters) after t seconds is given by the function $H(t) = 4.4t - 4.9t^2$.

i. Find $H'(t)$.

---

ii. At what rate is $H(t)$ changing after 1 second?

---

iii. At what time is $H'(t) = 0$? What is the significance of this time?

---

iv. When does the flea "land" (return to the initial height)? At what rate is $H(t)$ changing at this time? Is it increasing or decreasing?

Textbook Problems

40 Find the rate of change of a population of size $P(t) = t^3 + 4t + 1$ at time $t = 2$.

---

37 Let $P(q) = 6q - q^2$.

(a) Use a graph of $P(q)$ to determine whether each of the following derivatives is positive, negative, or zero: $P'(1)$, $P'(3)$, $P'(4)$. Explain.

(b) Find $P'(q)$ and the three derivatives in part (a).

39 Let $f(t) = t^2 - 4t + 5$.

(a) Find $f'(t)$.

(b) Find $f'(1)$ and $f'(2)$.

(c) Use a graph of $f(t)$ to check that your answers to part (b) are reasonable. Explain.

61 Let $f(x) = x^3 - 6x^2 - 15x + 20$. Find $f'(x)$ and all values of x for which $f'(x) = 0$. Explain the relationship between these values of x and the graph of $f(x)$.

41 The height of a sand dune (in centimeters) is represented by $f(t) = 700 - 3t^2$, where t is measured in years since 2005. Find $f(5)$ and $f'(5)$. Using units, explain what each means in terms of the sand dune.

42 Zebra mussels are freshwater shellfish that first appeared in the St. Lawrence River in the early 1980s and have spread throughout the Great Lakes. Suppose that t months after they appeared in a small bay, the number of zebra mussels is given by $Z(t) = 300t^2$. How many zebra mussels are in the bay after four months? At what rate is the population growing at that time? Give units.

43 The quantity, Q, in tons, of material at a municipal waste site is a function of the number of years since 2000, with $Q = f(t) = 3t^2 + 100$. Find $f(10)$, $f'(10)$, and the relative rate of change f'/f at $t = 10$. Interpret your answers in terms of waste.

44 The number, N, of acres of harvested land in a region is given by $N = f(t) = 120\sqrt{t}$, where t is the number of years since farming began in the region. Find $f(9)$, $f'(9)$, and the relative rate of change f'/f at $t = 9$. Interpret your answers in terms of harvested land.

47 The time, T, in seconds for one complete oscillation of a pendulum is given by $T = f(L) = 1.111\sqrt{L}$, where L is the length of the pendulum in feet. Find the following quantities, with units, and interpret in terms of the pendulum.

(a) $f(100)$

(b) $f'(100)$

48 Kleiber's Law states that the daily calorie requirement, $C(w)$, of a mammal is proportional to the mammal's body weight w raised to the 0.75 power. If body weight is measured in pounds, the constant of proportionality is approximately 42.

(a) Give formulas for $C(w)$ and $C'(w)$.

(b) Find and interpret (i) $C(10)$ and $C'(10)$ (ii) $C(100)$ and $C'(100)$ (iii) $C(1000)$ and $C'(1000)$

52 If you are outdoors, the wind may make it feel a lot colder than the thermometer reads. You feel the windchill temperature, which, if the air temperature is 20°F, is given in $^\circ$F by $W(v) = 48.17 - 27.2v^{0.16}$, where v is the wind velocity in mph for $5 \leq v \leq 60$.

(a) If the air temperature is 20°F, and the wind is blowing at 40 mph, what is the windchill temperature, to the nearest degree?

(b) Find $W'(40)$, and explain what this means in terms of windchill.

53

(a) Use the formula for the area of a circle of radius r, $A = \pi r^2$, to find dA/dr.

(b) The result from part (a) should look familiar. What does dA/dr represent geometrically?

(c) Use the difference quotient to explain the observation you made in part (b).

55 The cost to produce q items is $C(q) = 1000 + 2q^2$ dollars. Find the marginal cost of producing the 25th item. Interpret your answer in terms of costs.

56 The demand curve for a product is given by $q = 300 - 3p$, where p is the price of the product and q is the quantity that consumers buy at this price.

(a) Write the revenue as a function, $R(p)$, of price.

(b) Find $R'(10)$ and interpret your answer in terms of revenue.

(c) For what prices is $R'(p)$ positive? For what prices is it negative?

57 The yield, Y, of an apple orchard (measured in bushels of apples per acre) is a function of the amount x of fertilizer in pounds used per acre. Suppose $Y = f(x) = 320 + 140x - 10x^2$.

(a) What is the yield if 5 pounds of fertilizer is used per acre?

(b) Find $f'(5)$. Give units with your answer and interpret it in terms of apples and fertilizer.

(c) Given your answer to part (b), should more or less fertilizer be used? Explain.

58 The demand for a product is given, for $p, q \geq 0$, by $p = f(q) = 50 - 0.03q^2$.

(a) Find the p- and q-intercepts for this function and interpret them in terms of demand for this product.

(b) Find $f(20)$ and give units with your answer. Explain what it tells you in terms of demand.

(c) Find $f'(20)$ and give units with your answer. Explain what it tells you in terms of demand.

59 The cost (in dollars) of producing q items is given by $C(q) = 0.08q^3 + 75q + 1000$.

(a) Find the marginal cost function.

(b) Find $C(50)$ and $C'(50)$. Give units with your answers and explain what each is telling you about costs of production.

60 A ball is dropped from the top of the Empire State Building. The height, y, of the ball above the ground (in feet) is given as a function of time, t, (in seconds) by $y = 1250 - 16t^2$.

(a) Find the velocity of the ball at time t. What is the sign of the velocity? Why is this to be expected?

(b) When does the ball hit the ground, and how fast is it going at that time? Give your answer in feet per second and in miles per hour (1 ft/sec $= 15/22$ mph).

63 If the demand curve is a line, we can write $p = b + mq$, where p is the price of the product, q is the quantity sold at that price, and b and m are constants.

(a) Write the revenue as a function of quantity sold.

(b) Find the marginal revenue function.

3.1.6 The Second Derivative

Concavity $f'' > 0 \Rightarrow$ concave up $\cup$ $f'' < 0 \Rightarrow$ concave down $\cap$

Lecture Problems

Ex 10 Find and interpret the second derivatives of

i. $f(x) = x^2$

ii. $g(x) = x^3$

Textbook Problems

45 If $f(t) = 2t^3 - 4t^2 + 3t - 1$, find $f'(t)$ and $f''(t)$.

46 If $f(t) = t^4 - 3t^2 + 5t$, find $f'(t)$ and $f''(t)$.

Answers · Section 3.1

3.1.1 Constant, Linear and Constant Multiple

Ex 1 $\dfrac{d}{dx}(5) = 0$ (derivative of a constant is 0).

Ex 2 $\dfrac{d}{dx}(2x+3) = 2$ (linear function: derivative = slope).

Ex 3 $f'(x) = 7$

$\dfrac{d}{dx}\big(2f(x)\big) = 2\dfrac{d}{dx}\big(f(x)\big) = 2f'(x) = 2(7) = 14$

Another solution: $\dfrac{d}{dx}\big(2(5+7x)\big) = \dfrac{d}{dx}(10+14x) = 14$

1 $y' = 0$

2 $y' = 3$

8 $y' = 5$

29 $f'(x) = 2kx$

35 $\dfrac{dw}{dq} = 3ab^2$

36 $h(x) = \frac{a}{c}x + \frac{b}{c} \Rightarrow h'(x) = \dfrac{a}{c}$

3.1.2 The Power Rule

Ex 4 i. $3x^{3-1} = 3x^2$ ii. $-5x^{-5-1} = -5x^{-6}$ iii. $\frac{3}{2}x^{\frac{3}{2}-1} = \frac{3}{2}x^{1/2}$

Ex 5 i. $f'(x) = 8x^{8-1} = 8x^7$

ii. $h'(t) = 7t^{7-1} = 7t^6$

iii. $g(x) = x^{1/2} \Rightarrow g'(x) = \frac{1}{2}x^{\frac{1}{2}-1} = \frac{1}{2}x^{-1/2} = \dfrac{1}{2\sqrt{x}}$

iv. $w(t) = t^{-3} \Rightarrow w'(t) = -3t^{-3-1} = -3t^{-4} = -\dfrac{3}{t^4}$

v. $R'(x) = 2(4.5)x^{4.5-1} = 9x^{3.5}$

3 $y' = 12x^{11}$

4 $y' = -12x^{-13}$

5 $y' = \frac{4}{3}x^{1/3}$

6 $y' = 24t^2$

9 $f(x) = x^{-4} \Rightarrow f'(x) = -4x^{-5} = -\dfrac{4}{x^5}$

17 $g(t) = t^{-5} \Rightarrow g'(t) = -5t^{-6}$

18 $f(z) = -z^{-6.1} \Rightarrow f'(z) = 6.1z^{-7.1}$

19 $y = r^{-7/2} \Rightarrow y' = -\frac{7}{2}r^{-9/2}$

20 $y = x^{1/2} \Rightarrow y' = \frac{1}{2}x^{-1/2} = \dfrac{1}{2\sqrt{x}}$

21 $h = \theta^{-1/3} \Rightarrow h'(\theta) = -\frac{1}{3}\theta^{-4/3}$

22 $f(x) = x^{-3/2} \Rightarrow f'(x) = -\frac{3}{2}x^{-5/2}$

62 $f'(x) = nx^{n-1} \Rightarrow f'(1) = n(1)^{n-1} = n$ (since 1 to any power is 1).

3.1.3 Derivatives of Polynomials

Ex 6 i. $r'(p) = 5p^4 + 3p^2$

ii. $f'(x) = 5(2)x^{2-1} - 7(3)x^{3-1} = 10x - 21x^2$

iii. $g(t) = \frac{1}{4}t^2 + 3 \Rightarrow g'(t) = \frac{1}{4}(2)t^{2-1} + 0 = \frac{1}{2}t = \dfrac{t}{2}$

7 $y' = 12t^3 - 4t$

10 $f'(q) = 3q^2$

11 $y' = 2x + 5$

12 $y' = 18x^2 + 8x - 2$

13 $y' = 6x + 7$

14 $y' = 24t^2 - 8t + 12$

15 $y' = 8.4q - 0.5$

16 $y' = -12x^3 - 12x^2 - 6$

23 $y = 3t^5 - 5t^{1/2} + 7t^{-1} \Rightarrow y' = 15t^4 - \frac{5}{2}t^{-1/2} - 7t^{-2}$

24 $y = z^2 + \frac{1}{2}z^{-1} \Rightarrow y' = 2z - \frac{1}{2}z^{-2}$

25 $y = 3t^2 + 12t^{-1/2} - t^{-2} \Rightarrow y' = 6t - 6t^{-3/2} + 2t^{-3}$

26 $h = 3t^{-1} + 4t^{-2} \Rightarrow h'(t) = -3t^{-2} - 8t^{-3}$

27 $y = x^{3/2} + x^{1/2} \Rightarrow y' = \frac{3}{2}x^{1/2} + \frac{1}{2}x^{-1/2}$

28 $h = \theta^{1/2} - \theta^{-1} \Rightarrow h'(\theta) = \frac{1}{2}\theta^{-1/2} + \theta^{-2}$

30 $y' = 2ax + b$

31 $\dfrac{dQ}{dP} = 2aP + 3bP^2$

32 $v = at^2 + bt^{-2} \Rightarrow \dfrac{dv}{dt} = 2at - 2bt^{-3}$

33 $P = a + bt^{1/2} \Rightarrow \dfrac{dP}{dt} = \frac{b}{2}t^{-1/2}$

34 $\dfrac{dV}{dr} = \frac{8}{3}\pi rb$

54 $W = kr^3 \Rightarrow \dfrac{dW}{dr} = 3kr^2$, so dW/dr is proportional to r^2.

38 $f'(x) = 3x^2 - 8x + 7$

$f'(0) = 7, \quad f'(2) = 3(4) - 8(2) + 7 = 3, \quad f'(-1) = 3(1) + 8 + 7 = 18$

3.1.4 Equation of the Tangent Line

Ex 7 **Find y_1 at $x_1 = 1$:** $y = (1)^3 + 2(1)^2 - 5(1) + 7 = 5 \Rightarrow P(x_1, y_1) = (1, 5)$

Find slope m: $y' = 3x^2 + 4x - 5 \Rightarrow m = y'(1) = 3(1)^2 + 4(1) - 5 = 2$

Eq. of tangent: $y - y_1 = m(x - x_1) \Rightarrow y - 5 = 2(x - 1) \Rightarrow y - 5 = 2x - 2$

$\boxed{y = 2x + 3}$

49 $P(x_1, y_1) = (1, 1) \quad f'(x) = 6x^2 - 4x \Rightarrow m = f'(1) = 2$

$y - 1 = 2(x - 1) \Rightarrow \boxed{y = 2x - 1}$

50 (a) $y_1 = f(2) = 8$, $f'(x) = 3x^2 \Rightarrow m = f'(2) = 12$. $y - 8 = 12(x - 2) \Rightarrow \boxed{y = 12x - 16}$

(b) Near $x = 2$ the graph of x^3 is concave up, so it lies above the tangent line. The estimates are **underestimates**.

51 $f(4) = 24 - 16 = 8 \Rightarrow P = (4, 8)$ $f'(t) = 6 - 2t \Rightarrow m = f'(4) = -2$
$y - 8 = -2(t - 4) \Rightarrow \boxed{y = -2t + 16}$ (parabola opening down, vertex at $t = 3$; the tangent line touches it at $(4, 8)$ and lies above it).

3.1.5 Rate of Change and Applications

Ex 8 ARC $= P'(2)$: $P'(t) = 3t^2 + 4 \Rightarrow P'(2) = 3(2)^2 + 4 = 16$
Relative rate of change: $\text{RRC} = \dfrac{P'(2)}{P(2)} = \dfrac{16}{(2)^3 + 4(2) + 1} = \dfrac{16}{17} \times 100\% \approx 94.1\%$

Ex 9 i. $H'(t) = 4.4 - 4.9(2)t \Rightarrow H'(t) = 4.4 - 9.8t$
ii. $H'(1) = 4.4 - 9.8(1) = -5.4$ m/s
iii. $H'(t) = 0 \Rightarrow 4.4 - 9.8t = 0 \Rightarrow -9.8t = -4.4 \Rightarrow t \approx 0.45$ seconds (critical point). At $t = 0.45$ the flea reaches its maximum height, then the height starts to decrease ($H' > 0$ before, $H' < 0$ after).
iv. $H(t) = 0 \Rightarrow 4.4t - 4.9t^2 = 0 \Rightarrow t(4.4 - 4.9t) = 0 \Rightarrow t = 0$ or $t = \dfrac{4.4}{4.9} \approx 0.898$ seconds.
$H'(0.898) = 4.4 - 9.8(0.898) \approx -4.4$ m/s. The height is decreasing at 0.898 seconds.

40 $P'(t) = 3t^2 + 4 \Rightarrow P'(2) = 3(2)^2 + 4 = 16$. *(Same as lecture Ex 8.)*

37 (a) The graph is a parabola opening down with vertex at $q = 3$. At $q = 1$ the graph is increasing $\Rightarrow P'(1) > 0$; at $q = 3$ it is the top (horizontal tangent) $\Rightarrow P'(3) = 0$; at $q = 4$ it is decreasing $\Rightarrow P'(4) < 0$.
(b) $P'(q) = 6 - 2q$: $P'(1) = 4$, $P'(3) = 0$, $P'(4) = -2$.

39 (a) $f'(t) = 2t - 4$ (b) $f'(1) = -2$, $f'(2) = 0$
(c) The graph is a parabola opening up with vertex at $t = 2$. At $t = 1$ the graph is decreasing (negative slope) and at $t = 2$ it is at the bottom (horizontal tangent, slope 0). So the answers are reasonable.

61 $f'(x) = 3x^2 - 12x - 15 = 3(x^2 - 4x - 5) = 3(x - 5)(x + 1)$
$f'(x) = 0 \Rightarrow x = -1$ or $x = 5$. At these values the tangent line is horizontal: f has a local maximum at $x = -1$ and a local minimum at $x = 5$.

41 $f(5) = 700 - 3(25) = 625$ cm $\quad f'(t) = -6t \Rightarrow f'(5) = -30$ cm/year
In 2010 the sand dune is 625 cm high, and its height is decreasing at a rate of 30 cm per year.

42 $Z(4) = 300(4)^2 = 4800$ zebra mussels.
$Z'(t) = 600t \Rightarrow Z'(4) = 2400$ zebra mussels per month. After 4 months the population is growing by about 2400 mussels per month.

43 $f(10) = 3(100) + 100 = 400$ tons $\quad f'(t) = 6t \Rightarrow f'(10) = 60$ tons/year
$\dfrac{f'(10)}{f(10)} = \dfrac{60}{400} = 0.15 = 15\%$ per year
In 2010 there are 400 tons of waste at the site, increasing at 60 tons per year, which is 15% per year.

44 $f(9) = 120\sqrt{9} = 360$ acres $\quad f(t) = 120t^{1/2} \Rightarrow f'(t) = 60t^{-1/2} \Rightarrow f'(9) = \dfrac{60}{3} = 20$ acres/year

$\dfrac{f'(9)}{f(9)} = \dfrac{20}{360} \approx 0.0556 = 5.56\%$ per year

After 9 years, 360 acres are harvested, increasing by 20 acres per year, which is about 5.56% per year.

47 (a) $f(100) = 1.111\sqrt{100} = 11.11$ seconds: a pendulum 100 feet long takes 11.11 seconds for one complete oscillation.

(b) $f(L) = 1.111L^{1/2} \Rightarrow f'(L) = 0.5555L^{-1/2} \Rightarrow f'(100) = \dfrac{0.5555}{10} \approx 0.056$ seconds per foot: increasing the length from 100 ft by 1 ft increases the time by about 0.056 seconds.

48 (a) $C(w) = 42w^{0.75} \quad C'(w) = 42(0.75)w^{-0.25} = 31.5w^{-0.25}$

(b) (i) $C(10) \approx 236$ calories; $C'(10) \approx 17.7$ calories per pound

(ii) $C(100) \approx 1328$ calories; $C'(100) \approx 9.96$ calories per pound

(iii) $C(1000) \approx 7469$ calories; $C'(1000) \approx 5.60$ calories per pound

A 10-lb mammal needs about 236 calories a day, and each extra pound adds about 17.7 calories. Heavier mammals need more calories, but each extra pound adds fewer calories.

52 (a) $W(40) = 48.17 - 27.2(40)^{0.16} \approx 48.17 - 49.08 \approx -1^\circ\text{F}$

(b) $W'(v) = -27.2(0.16)v^{-0.84} = -4.352v^{-0.84} \Rightarrow W'(40) \approx -0.20$ °F per mph. When the wind is 40 mph, each extra 1 mph of wind makes it feel about 0.2°F colder.

53 (a) $\dfrac{dA}{dr} = 2\pi r$ (b) $2\pi r$ is the **circumference** of the circle.

(c) $\dfrac{A(r+h) - A(r)}{h}$: when the radius grows by a small h, the extra area is a thin ring of length about $2\pi r$ and width h, so the extra area $\approx 2\pi r \cdot h$ and the quotient $\approx 2\pi r$.

55 $C'(q) = 4q \Rightarrow C'(25) = 100$ dollars per item. Producing the 25th item costs about \$100.

56 (a) $R(p) = p \cdot q = p(300 - 3p) = 300p - 3p^2$

(b) $R'(p) = 300 - 6p \Rightarrow R'(10) = 240$. When the price is \$10, raising the price by \$1 increases revenue by about \$240.

(c) $R'(p) > 0$ when $p < 50$; $R'(p) < 0$ when $50 < p \le 100$ (since $q \ge 0$ means $p \le 100$).

57 (a) $f(5) = 320 + 700 - 250 = 770$ bushels per acre

(b) $f'(x) = 140 - 20x \Rightarrow f'(5) = 40$ bushels per acre per pound of fertilizer: using 1 more pound increases the yield by about 40 bushels per acre.

(c) Since $f'(5) > 0$, the yield is still increasing, so **more** fertilizer should be used.

58 (a) p-intercept: $q = 0 \Rightarrow p = 50$: at \$50 nobody buys the product. q-intercept: $50 - 0.03q^2 = 0 \Rightarrow q^2 \approx 1666.7 \Rightarrow q \approx 40.8$: about 41 items are wanted if the product is free.

(b) $f(20) = 50 - 0.03(400) = 38$ dollars: to sell 20 items, the price must be \$38.

(c) $f'(q) = -0.06q \Rightarrow f'(20) = -1.2$ dollars per item: to sell one more item (the 21st), the price must drop by about \$1.20.

59 (a) $C'(q) = 0.24q^2 + 75$

(b) $C(50) = 0.08(125000) + 3750 + 1000 = 14{,}750$ dollars: producing 50 items costs \$14,750.

$C'(50) = 0.24(2500) + 75 = 675$ dollars per item: the 51st item costs about \$675.

60 (a) $v = y' = -32t$ ft/s. The velocity is negative because the height is decreasing (the ball is falling).
(b) $y = 0 \Rightarrow 16t^2 = 1250 \Rightarrow t^2 = 78.125 \Rightarrow t \approx 8.84$ seconds.
$v(8.84) = -32(8.84) \approx -282.8$ ft/s, so the speed is about 282.8 ft/s $\approx 282.8 \times \frac{15}{22} \approx 192.8$ mph.

63 (a) $R(q) = p \cdot q = (b + mq)q = bq + mq^2$ (b) $R'(q) = b + 2mq$

3.1.6 The Second Derivative

Ex 10 i. $f'(x) = 2x,\ f''(x) = 2$. $f''(x)$ is positive for all x on $(-\infty, \infty) \Rightarrow f(x)$ is concave up on $(-\infty, \infty)$.
ii. $g'(x) = 3x^2,\ g''(x) = 3(2x) = 6x$. $g''(x) = 0 \Rightarrow x = 0$.
Test: $g''(-1) = -6 < 0$ (concave down $\cap$), $g''(1) = 6 > 0$ (concave up $\cup$).
Since $g''(x)$ is negative on $(-\infty, 0)$ and positive on $(0, \infty)$, $g(x)$ is concave down on $(-\infty, 0)$ and concave up on $(0, \infty)$.

45 $f'(t) = 6t^2 - 8t + 3 \quad f''(t) = 12t - 8$

46 $f'(t) = 4t^3 - 6t + 5 \quad f''(t) = 12t^2 - 6$

3.2 Exponential and Logarithmic Functions

In this section

1. The Exponential Rule 2 lecture · 22 textbook
2. The Derivative of ln x 1 lecture · 6 textbook
3. Equation of the Tangent Line 1 lecture · 7 textbook
4. Rate of Change and Applications 2 lecture · 15 textbook

3.2.1 The Exponential Rule

Exponential rule $\dfrac{d}{dx}\left(a^x\right) = (\ln a)\, a^x$ and $\dfrac{d}{dx}\left(a^{kx}\right) = k(\ln a)\, a^{kx}$

Base e $\dfrac{d}{dx}\left(e^x\right) = e^x$ and $\dfrac{d}{dx}\left(e^{kx}\right) = ke^{kx}$

Lecture Problems

Ex 1 Differentiate $y = 2.3^x + 5e^x$.

Ex 2 Find the derivative of $P = 5 + 3x^2 - 7e^{-0.2x}$.

Textbook Problems

Differentiate. Assume A, B, and C are constants.

1 $f(x) = 2e^x + x^2$

2 $P = 3t^3 + 2e^t$

3 $y = 5t^2 + 4e^t$

4 $f(x) = x^3 + 3^x$

5 $y = 2^x + \dfrac{2}{x^3}$

6 $y = 5 \cdot 5^t + 6 \cdot 6^t$

7 $f(x) = 2^x + 2 \cdot 3^x$

8 $y = 4 \cdot 10^x - x^3$

9 $y = 3x - 2 \cdot 4^x$

10 $y = 5 \cdot 2^x - 5x + 4$

11 $f(t) = e^{3t}$

12 $y = e^{0.7t}$

13 $y = e^{-4t}$

14 $P = e^{-0.2t}$

15 $P = 50e^{-0.6t}$

16 $P = 200e^{0.12t}$

17 $P(t) = 3000(1.02)^t$

18 $P(t) = 12.41(0.94)^t$

19 $P(t) = Ce^t$

20 $y = B + Ae^t$

21 $f(x) = Ae^x - Bx^2 + C$

22 $y = 10^x + \dfrac{10}{x}$

3.2.2 The Derivative of ln x

Natural log $\dfrac{d}{dx}(\ln x) = \dfrac{1}{x}$

Lecture Problems

Ex 3 Differentiate $y = 5\ln t + 7e^t - 4t^2 + 12$.

Textbook Problems

Differentiate. Assume A, B, and C are constants.

23 $R = 3\ln q$

24 $D = 10 - \ln p$

25 $y = t^2 + 5\ln t$

26 $R(q) = q^2 - 2\ln q$

27 $y = x^2 + 4x - 3\ln x$

28 $f(t) = Ae^t + B\ln t$

3.2.3 Equation of the Tangent Line

Tangent line $y - y_1 = m(x - x_1)$, where $m = f'(x_1)$

Lecture Problems

Ex 6 Find the equation of the tangent line to the graph of $f(x) = \ln x$ at the point where $x = 2$.

Textbook Problems

29 For $f(t) = 4 - 2e^t$, find $f'(-1)$, $f'(0)$, and $f'(1)$. Graph $f(t)$, and draw tangent lines at $t = -1$, $t = 0$, and $t = 1$. Do the slopes of the lines match the derivatives you found?

30 Find the equation of the tangent line to the graph of $y = 3^x$ at $x = 1$. Check your work by sketching a graph of the function and the tangent line on the same axes.

31 Find the equation of the tangent line to $y = e^{-2t}$ at $t = 0$. Check by sketching the graphs of $y = e^{-2t}$ and the tangent line on the same axes.

32 Find the equation of the tangent line to $f(x) = 10e^{-0.2x}$ at $x = 4$.

41 Find the value of c where the line l tangent to the graph of $y = 2^x$ at $(0, 1)$ intersects the x-axis.

49

(a) Find the equation of the tangent line to $y = \ln x$ at $x = 1$.

(b) Use it to calculate approximate values for $\ln(1.1)$ and $\ln(2)$.

(c) Using a graph, explain whether the approximate values are smaller or larger than the true values. Would the same result have held if you had used the tangent line to estimate $\ln(0.9)$ and $\ln(0.5)$? Why?

50 Find the quadratic polynomial $g(x) = ax^2 + bx + c$ which best fits the function $f(x) = e^x$ at $x = 0$, in the sense that $g(0) = f(0)$, and $g'(0) = f'(0)$, and $g''(0) = f''(0)$. Using a computer or calculator, sketch graphs of f and g on the same axes. What do you notice?

3.2.4 Rate of Change and Applications

Rate of change	Rate of change at $t = a$: $f'(a)$
Relative rate	Relative rate of change at $t = a$: $\dfrac{f'(a)}{f(a)} \times 100\%$

Lecture Problems

Ex 4 The population of Nevada, P, in millions, can be approximated by $P = 2.020(1.036)^t$, where t is years since the start of 2000.

i. At what rate was the population growing at the beginning of 2009? Give units with your answer.

ii. What is the relative rate of change (RRC) of the population of Nevada at the beginning of 2009?

Ex 5 It is projected that t years from now, the population of a certain country will become $P(t) = 50e^{0.02t}$ million.

i. At what rate will the population be changing (with respect to time) 10 years from now?

ii. At what percentage (i.e., RRC) will the population be changing (with respect to time) t years from now? Does this percentage rate depend on t or is it constant?

Textbook Problems

33 A fish population is approximated by $P(t) = 10e^{0.6t}$, where t is in months. Calculate and use units to explain what each of the following tells us about the population:

(a) $P(12)$

(b) $P'(12)$

34 The world's population is about $f(t) = 6.8e^{0.012t}$ billion, where t is time in years since 2009. Find $f(0)$, $f'(0)$, $f(10)$, and $f'(10)$. Using units, interpret your answers in terms of population.

35 The demand curve for a product is given by $q = f(p) = 10{,}000e^{-0.25p}$, where q is the quantity sold and p is the price of the product, in dollars. Find $f(2)$ and $f'(2)$. Explain in economic terms what information each of these answers gives you.

36 Worldwide production of solar power, in megawatts, can be modeled by $f(t) = 1040(1.3)^t$, where t is years since 2000. Find $f(0)$, $f'(0)$, $f(15)$, and $f'(15)$. Give units and interpret your answers in terms of solar power.

37 A new DVD is available for sale in a store one week after its release. The cumulative revenue, $\$R$, from sales of the DVD in this store in week t after its release is $R = f(t) = 350 \ln t$ with $t > 1$. Find $f(5)$, $f'(5)$, and the relative rate of change f'/f at $t = 5$. Interpret your answers in terms of revenue.

38 In 2009, the population of Hungary was approximated by $P = 9.906(0.997)^t$, where P is in millions and t is in years since 2009. Assume the trend continues.

(a) What does this model predict for the population of Hungary in the year 2020?

(b) How fast (in people/year) does this model predict Hungary's population will be decreasing in 2020?

39 With t in years since January 1, 2010, the population P of Slim Chance is predicted by $P = 35{,}000(0.98)^t$. At what rate will the population be changing on January 1, 2023?

40 Some antique furniture increased very rapidly in price over the past decade. For example, the price of a particular rocking chair is well approximated by $V = 75(1.35)^t$, where V is in dollars and t is in years since 2000. Find the rate, in dollars per year, at which the price is increasing at time t.

42 At a time t hours after it was administered, the concentration of a drug in the body is $f(t) = 27e^{-0.14t}$ ng/ml. What is the concentration 4 hours after it was administered? At what rate is the concentration changing at that time?

43 The cost of producing a quantity, q, of a product is given by $C(q) = 1000 + 30e^{0.05q}$ dollars. Find the cost and the marginal cost when $q = 50$. Interpret these answers in economic terms.

44 With time, t, in minutes, the temperature, H, in degrees Celsius, of a bottle of water put in the refrigerator at $t = 0$ is given by $H = 4 + 16e^{-0.02t}$. How fast is the water cooling initially? After 10 minutes? Give units.

45 Carbon-14 is a radioactive isotope used to date objects. If A_0 represents the initial amount of carbon-14 in the object, then the quantity remaining at time t, in years, is $A(t) = A_0e^{-0.000121t}$.

(a) A tree, originally containing 185 micrograms of carbon-14, is now 500 years old. At what rate is the carbon-14 decaying?

(b) In 1988, scientists found that the Shroud of Turin, which was reputed to be the burial cloth of Jesus, contained 91% of the amount of carbon-14 in freshly made cloth of the same material. According to this data, how old was the Shroud of Turin in 1988?

46 For the cost function $C = 1000 + 300 \ln q$ (in dollars), find the cost and the marginal cost at a production level of 500. Interpret your answers in economic terms.

47 In 2009, the population, P, of India was 1.166 billion and growing at 1.5% annually.

(a) Give a formula for P in terms of time, t, measured in years since 2009.

(b) Find $\dfrac{dP}{dt}$, $\left.\dfrac{dP}{dt}\right|_{t=0}$, and $\left.\dfrac{dP}{dt}\right|_{t=25}$. What do each of these represent in practical terms?

48 In 2009, the population of Mexico was 111 million and growing 1.13% annually, while the population of the US was 307 million and growing 0.975% annually. If we measure growth rates in people/year, which population was growing faster in 2009?

Answers · Section 3.2

3.2.1 The Exponential Rule

Ex 1 $a = 2.3$: $y' = \ln(2.3)\,2.3^x + 5e^x$

Ex 2 $e^{-0.2x}$: $k = -0.2$

$P' = 0 + 3(2x) - 7(-0.2)e^{-0.2x}$

$\boxed{P' = 6x + 1.4e^{-0.2x}}$

1 $f'(x) = 2e^x + 2x$

2 $P' = 9t^2 + 2e^t$

3 $y' = 10t + 4e^t$

4 $f'(x) = 3x^2 + (\ln 3)3^x$

5 $y = 2^x + 2x^{-3} \Rightarrow y' = (\ln 2)2^x - 6x^{-4}$

6 $y' = 5(\ln 5)5^t + 6(\ln 6)6^t$

7 $f'(x) = (\ln 2)2^x + 2(\ln 3)3^x$

8 $y' = 4(\ln 10)10^x - 3x^2$

9 $y' = 3 - 2(\ln 4)4^x$

10 $y' = 5(\ln 2)2^x - 5$

11 $f'(t) = 3e^{3t}$

12 $y' = 0.7e^{0.7t}$

13 $y' = -4e^{-4t}$

14 $P' = -0.2e^{-0.2t}$

15 $P' = 50(-0.6)e^{-0.6t} = -30e^{-0.6t}$

16 $P' = 200(0.12)e^{0.12t} = 24e^{0.12t}$

17 $P'(t) = 3000\ln(1.02)(1.02)^t \approx 59.41(1.02)^t$

18 $P'(t) = 12.41\ln(0.94)(0.94)^t \approx -0.768(0.94)^t$

19 $P'(t) = Ce^t$

20 $y' = Ae^t$

21 $f'(x) = Ae^x - 2Bx$

22 $y = 10^x + 10x^{-1} \Rightarrow y' = (\ln 10)10^x - \dfrac{10}{x^2}$

3.2.2 The Derivative of ln x

Ex 3 $y' = 5\left(\dfrac{1}{t}\right) + 7e^t - 4(2t) + 0$

$\boxed{y' = \dfrac{5}{t} + 7e^t - 8t}$

23 $R' = \dfrac{3}{q}$

24 $D' = -\dfrac{1}{p}$

25 $y' = 2t + \dfrac{5}{t}$

26 $R'(q) = 2q - \dfrac{2}{q}$

27 $y' = 2x + 4 - \dfrac{3}{x}$

28 $f'(t) = Ae^t + \dfrac{B}{t}$

3.2.3 Equation of the Tangent Line

Ex 6 **Find y_1 when $x_1 = 2$:** $y_1 = f(2) = \ln(2) \Rightarrow (x_1, y_1) = (2, \ln 2)$
Find slope m: $f'(x) = \dfrac{1}{x} \Rightarrow m = f'(2) = \dfrac{1}{2}$
Eq. of tangent: $y - \ln(2) = \frac{1}{2}(x - 2) \Rightarrow y - \ln(2) = \frac{1}{2}x - 1 \Rightarrow y = \frac{1}{2}x - 1 + \ln(2)$
$\boxed{y = \frac{1}{2}x - 0.307}$

29 $f'(t) = -2e^t$: $f'(-1) = -2e^{-1} \approx -0.736$, $f'(0) = -2$, $f'(1) = -2e \approx -5.437$
The graph is decreasing and gets steeper as t increases, so all three tangent lines have negative slopes that get steeper. Yes, the slopes match.

30 $y_1 = 3^1 = 3 \Rightarrow (1, 3)$ $y' = (\ln 3)3^x \Rightarrow m = 3\ln 3 \approx 3.296$
$y - 3 = 3.296(x - 1) \Rightarrow \boxed{y = 3.296x - 0.296}$

31 $y_1 = e^0 = 1 \Rightarrow (0, 1)$ $y' = -2e^{-2t} \Rightarrow m = -2$
$y - 1 = -2(t - 0) \Rightarrow \boxed{y = -2t + 1}$

32 $y_1 = f(4) = 10e^{-0.8} \approx 4.493 \Rightarrow (4,\ 4.493)$
$f'(x) = 10(-0.2)e^{-0.2x} = -2e^{-0.2x} \Rightarrow m = f'(4) = -2e^{-0.8} \approx -0.899$
$y - 4.493 = -0.899(x - 4) \Rightarrow \boxed{y = -0.899x + 8.088}$

41 $y' = (\ln 2)2^x \Rightarrow m = \ln 2$ at $x = 0$. Tangent: $y - 1 = (\ln 2)(x - 0) \Rightarrow y = 1 + (\ln 2)x$
At the x-axis $y = 0$: $0 = 1 + (\ln 2)c \Rightarrow c = -\dfrac{1}{\ln 2} \approx -1.443$

49 (a) $y_1 = \ln 1 = 0$, $y' = \dfrac{1}{x} \Rightarrow m = 1$. $y - 0 = 1(x - 1) \Rightarrow \boxed{y = x - 1}$
(b) $\ln(1.1) \approx 1.1 - 1 = 0.1$ $\ln(2) \approx 2 - 1 = 1$
(c) $\ln x$ is concave down, so its graph lies below the tangent line. The approximations are larger than the true values ($\ln 1.1 \approx 0.0953$, $\ln 2 \approx 0.693$). Yes, the same holds for $\ln(0.9) \approx -0.1$ and $\ln(0.5) \approx -0.5$ (true values -0.105 and -0.693), because the curve is below the tangent line on both sides of $x = 1$.

50 $f(0) = f'(0) = f''(0) = e^0 = 1$. $g'(x) = 2ax + b$, $g''(x) = 2a$
$g(0) = c = 1$, $g'(0) = b = 1$, $g''(0) = 2a = 1 \Rightarrow a = \frac{1}{2}$
$\boxed{g(x) = \frac{1}{2}x^2 + x + 1}$. The two graphs are very close near $x = 0$ and separate as x moves away from 0.

3.2.4 Rate of Change and Applications

Ex 4 In 2000, $t = 0$, so the beginning of 2009 is $t = 9$.
i. $P' = 2.02\ln(1.036)(1.036)^t$
$P'(9) = 2.02\ln(1.036)(1.036)^9 \approx 0.0982$ million/year
The population of Nevada was growing at a rate of about 0.0982 million, or 98,200, people per year at the start of 2009.
ii. $\text{RRC} = \dfrac{P'(9)}{P(9)} = \dfrac{0.0982}{2.020(1.036)^9} \times 100\% \approx 0.0354 \times 100\% = 3.54\%$ per year

Ex 5 i. $P'(t) = 50(0.02)e^{0.02t} = e^{0.02t}$ million/year
$P'(10) = e^{0.02(10)} \approx 1.221$ million/year
10 years from now, the population will be growing by 1.221 million per year.
ii. $\text{RRC} = \dfrac{P'(t)}{P(t)} = \dfrac{e^{0.02t}}{50e^{0.02t}} \times 100\% = \dfrac{1}{50} \times 100\% = 2\%$ per year
The percentage rate is constant and does not depend on t; it is the percentage of k (the continuous rate of change).

---

33 (a) $P(12) = 10e^{7.2} \approx 13{,}394$ fish: after 12 months there are about 13,394 fish.
(b) $P'(t) = 10(0.6)e^{0.6t} = 6e^{0.6t} \Rightarrow P'(12) = 6e^{7.2} \approx 8037$ fish per month: after 12 months the population is growing by about 8037 fish per month.

---

34 $f'(t) = 6.8(0.012)e^{0.012t} = 0.0816e^{0.012t}$
$f(0) = 6.8$ billion people in 2009; $f'(0) = 0.0816$ billion people per year: in 2009 the population grows by about 81.6 million people per year.
$f(10) = 6.8e^{0.12} \approx 7.667$ billion people in 2019; $f'(10) = 0.0816e^{0.12} \approx 0.092$ billion people per year: in 2019 it grows by about 92 million people per year.

---

35 $f(2) = 10{,}000e^{-0.5} \approx 6065$ items: at a price of 2 dollars, about 6065 items are sold.
$f'(p) = 10{,}000(-0.25)e^{-0.25p} = -2500e^{-0.25p} \Rightarrow f'(2) = -2500e^{-0.5} \approx -1516$ items per dollar: at a price of 2 dollars, raising the price by 1 dollar decreases sales by about 1516 items.

---

36 $f'(t) = 1040\ln(1.3)(1.3)^t$
$f(0) = 1040$ megawatts in 2000; $f'(0) = 1040\ln(1.3) \approx 272.9$ megawatts per year: production was growing by about 273 MW per year in 2000.
$f(15) = 1040(1.3)^{15} \approx 53{,}233$ megawatts in 2015; $f'(15) \approx 13{,}967$ megawatts per year in 2015.

---

37 $f(5) = 350\ln 5 \approx 563.30$ dollars: by week 5 the revenue is about 563.30 dollars.
$f'(t) = \dfrac{350}{t} \Rightarrow f'(5) = 70$ dollars per week: in week 5 revenue is increasing by about 70 dollars per week.
$\dfrac{f'(5)}{f(5)} = \dfrac{70}{563.30} \approx 0.124 = 12.4\%$ per week.

---

38 2020 means $t = 11$.
(a) $P(11) = 9.906(0.997)^{11} \approx 9.584$ million people.
(b) $P'(t) = 9.906\ln(0.997)(0.997)^t \Rightarrow P'(11) \approx -0.0288$ million people per year. The population is decreasing by about 28,800 people per year.

---

39 January 1, 2023 means $t = 13$.
$P'(t) = 35{,}000\ln(0.98)(0.98)^t \Rightarrow P'(13) \approx -544$ people per year. The population will be decreasing by about 544 people per year.

---

40 $\dfrac{dV}{dt} = 75\ln(1.35)(1.35)^t \approx 22.51(1.35)^t$ dollars per year.

---

42 $f(4) = 27e^{-0.56} \approx 15.42$ ng/ml
$f'(t) = 27(-0.14)e^{-0.14t} = -3.78e^{-0.14t} \Rightarrow f'(4) = -3.78e^{-0.56} \approx -2.16$ ng/ml per hour. After 4 hours the concentration is decreasing by about 2.16 ng/ml per hour.

---

43 $C(50) = 1000 + 30e^{2.5} \approx 1365.47$ dollars: producing 50 items costs about 1365.47 dollars.
$C'(q) = 30(0.05)e^{0.05q} = 1.5e^{0.05q} \Rightarrow C'(50) = 1.5e^{2.5} \approx 18.27$ dollars per item: the 51st item costs about 18.27 dollars.

44 $H'(t) = 16(-0.02)e^{-0.02t} = -0.32e^{-0.02t}$
$H'(0) = -0.32$ °C per minute: initially the water cools by 0.32°C per minute.
$H'(10) = -0.32e^{-0.2} \approx -0.262$ °C per minute: after 10 minutes it cools by about 0.26°C per minute.

45 (a) $A(t) = 185e^{-0.000121t} \Rightarrow A'(t) = 185(-0.000121)e^{-0.000121t}$
$A'(500) \approx -0.0211$ micrograms per year: the carbon-14 is decaying at about 0.021 micrograms per year.
(b) $0.91A_0 = A_0e^{-0.000121t} \Rightarrow \ln(0.91) = -0.000121t \Rightarrow t = \dfrac{\ln(0.91)}{-0.000121} \approx 779$ years old.

46 $C(500) = 1000 + 300\ln 500 \approx 2864.38$ dollars: producing 500 items costs about 2864.38 dollars.
$C'(q) = \dfrac{300}{q} \Rightarrow C'(500) = 0.60$ dollars per item: the 501st item costs about 0.60 dollars.

47 (a) $P = 1.166(1.015)^t$ billion
(b) $\dfrac{dP}{dt} = 1.166\ln(1.015)(1.015)^t \approx 0.01736(1.015)^t$ billion people per year: the rate of growth at time t.
$\left.\dfrac{dP}{dt}\right|_{t=0} \approx 0.0174$ billion per year: in 2009 the population grows by about 17.4 million people per year.
$\left.\dfrac{dP}{dt}\right|_{t=25} \approx 0.0252$ billion per year: in 2034 it grows by about 25.2 million people per year.

48 Mexico: $P = 111(1.0113)^t \Rightarrow P'(0) = 111\ln(1.0113) \approx 1.25$ million people per year.
US: $P = 307(1.00975)^t \Rightarrow P'(0) = 307\ln(1.00975) \approx 2.98$ million people per year.
The US population was growing faster in people per year (even though its percentage rate is smaller).

3.3 The Chain Rule

In this section

1. The Chain Rule for Powers 1 lecture · 15 textbook
2. The Chain Rule for Exponentials and Logarithms 2 lecture · 17 textbook
3. The Chain Rule with Values and Graphs 2 lecture · 8 textbook
4. Relative Rates and Logarithms 2 lecture · 13 textbook

3.3.1 The Chain Rule for Powers

Chain rule $\dfrac{dy}{dt} = \dfrac{dy}{dz} \cdot \dfrac{dz}{dt}$ or $\dfrac{d}{dt}\big(f(g(t))\big) = f'(g(t)) \cdot g'(t)$

Power $\dfrac{d}{dt}\Big(\big(f(t)\big)^n\Big) = n\big(f(t)\big)^{n-1} f'(t)$

Lecture Problems

Ex 1 Find the derivative of the function $y = (4t^2 + 1)^7$.

Textbook Problems

Find the derivative of the function.

1 $(4x^2 + 1)^7$

2 $f(x) = (x + 1)^{99}$

3 $R = (q^2 + 1)^4$

4 $w = (t^2 + 1)^{100}$

5 $w = (t^3 + 1)^{100}$

6 $w = (5r - 6)^3$

7 $y = \sqrt{s^3 + 1}$

9 $C = 12(3q^2 - 5)^3$

22 $Q = 100(t^2 + 5)^{0.5}$

24 $y = (5 + e^x)^2$

25 $P = (1 + \ln x)^{0.5}$

26 $\sqrt{e^x + 1}$

27 $f(x) = \sqrt{1 - x^2}$

28 $f(\theta) = (e^\theta + e^{-\theta})^{-1}$

39 Find the equation of the tangent line to $f(x) = (x - 1)^3$ at the point where $x = 2$.

3.3.2 The Chain Rule for Exponentials and Logarithms

Exponential	$\dfrac{d}{dt}\left(e^{f(t)}\right) = e^{f(t)} f'(t)$
Logarithm	$\dfrac{d}{dt}\big(\ln f(t)\big) = \dfrac{f'(t)}{f(t)}$

Lecture Problems

Ex 2 Find the derivative of each of the following:

i. $(3t^3 - t)^5$

ii. $\ln(q^2 + 1)$

Ex 3 Differentiate each of the following:

i. $5\ln(2t^2+3)$

ii. $\sqrt{1+2e^{5t}}$

iii. $\dfrac{1}{\sqrt[3]{5x^2+7}}$

Textbook Problems

Find the derivative of the function.

8 $y = 12 - 3x^2 + 2e^{3x}$

10 $f(x) = 6e^{5x} + e^{-x^2}$

11 $y = 5e^{5t+1}$

12 $w = e^{-3t^2}$

13 $w = e^{\sqrt{s}}$

14 $y = \ln(5t+1)$

15 $f(x) = \ln(1-x)$

16 $f(t) = \ln(t^2+1)$

17 $f(x) = \ln(1-e^{-x})$

18 $f(x) = \ln(e^x+1)$

19 $f(t) = 5\ln(5t+1)$

20 $g(t) = \ln(4t+9)$

21 $y = 5 + \ln(3t + 2)$

23 $y = 5x + \ln(x + 2)$

40 A firm estimates that the total revenue, R, received from the sale of q goods is given by $R = \ln(1 + 1000q^2)$. Calculate the marginal revenue when $q = 10$.

41 The distance, s, of a moving body from a fixed point is given as a function of time by $s = 20e^{t/2}$. Find the velocity, v, of the body as a function of t.

42 If you invest P dollars in a bank account at an annual interest rate of $r\%$, then after t years you will have B dollars, where $B = P\left(1 + \dfrac{r}{100}\right)^t$.

(a) Find dB/dt, assuming P and r are constant. In terms of money, what does dB/dt represent?

(b) Find dB/dr, assuming P and t are constant. In terms of money, what does dB/dr represent?

3.3.3 The Chain Rule with Values and Graphs

Chain rule $\dfrac{dy}{dt} = \dfrac{dy}{dz} \cdot \dfrac{dz}{dt}$ or $\dfrac{d}{dt}\big(f(g(t))\big) = f'(g(t)) \cdot g'(t)$

Lecture Problems

Ex 4 Let $h(x) = f(g(x))$ and $k(x) = g(f(x))$. Given that $f(1) = 1$, $f'(0) = 0$, $f'(1) = 3$, $g(1) = 0$, $g'(1) = 1$. Evaluate:

i. $h'(1)$

ii. $k'(1)$

Ex 5 Let $h(x) = f(g(x))$ and $k(x) = g(f(x))$. Use Figure 3.17 to estimate:

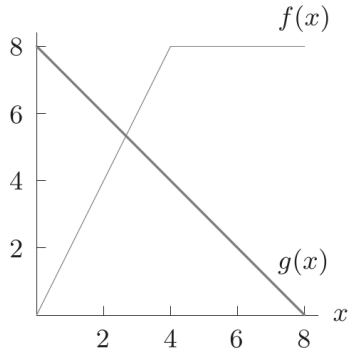


Figure 3.17

i. $h'(1)$

ii. $k'(2)$

Textbook Problems

44 Use Figure 3.18 to evaluate $\left.\frac{d}{dx} f(g(x))\right|_{x=30}$.

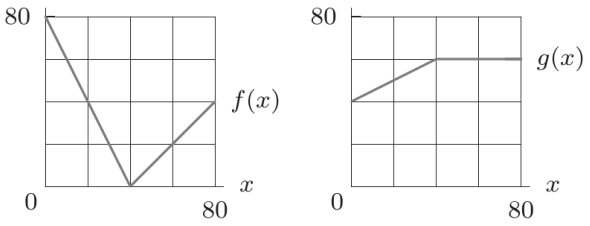


Figure 3.18

45 Use Figure 3.18 to evaluate $\left.\frac{d}{dx} f(g(x))\right|_{x=70}$.

46 Use Figure 3.18 to evaluate $\left.\frac{d}{dx} g(f(x))\right|_{x=30}$.

47 Use Figure 3.18 to evaluate $\left.\frac{d}{dx} g(f(x))\right|_{x=70}$.

48 Let $h(x) = f(g(x))$ and $k(x) = g(f(x))$. Use Figure 3.19 to estimate $h'(1)$.

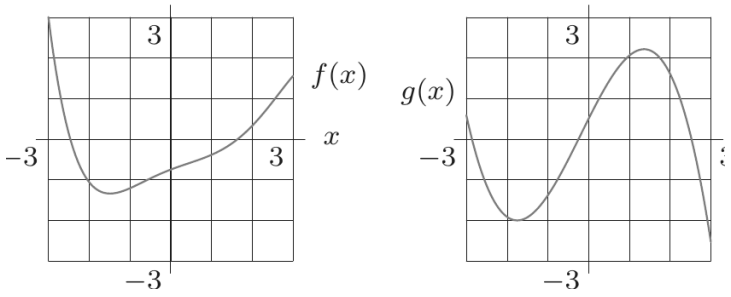


Figure 3.19

49 Use Figure 3.19 to estimate $k'(1)$.

50 Use Figure 3.19 to estimate $h'(2)$.

51 Use Figure 3.19 to estimate $k'(2)$.

3.3.4 Relative Rates and Logarithms

Relative rate Relative rate of change of $f(t) = \dfrac{f'(t)}{f(t)} = \dfrac{d}{dt}\big(\ln f(t)\big)$

Lecture Problems

Ex 6 Find the relative rate of change of the exponential function $z = P_0 e^{kt}$.

Ex 7 The surface area S of a mammal, in cm^2, is a function of the body mass, M, of the mammal, in kilograms, and is given by $S = 1095M^{2/3}$. Find the relative rate of change of S with respect to M and evaluate for a human with body mass 70 kilograms. Interpret your answer.

Textbook Problems

29 Find the relative rate of change, $f'(t)/f(t)$, of the function $f(t) = 10t + 5$.

30 Find the relative rate of change, $f'(t)/f(t)$, of the function $f(t) = 15t + 12$.

31 Find the relative rate of change, $f'(t)/f(t)$, of the function $f(t) = 8e^{5t}$.

32 Find the relative rate of change, $f'(t)/f(t)$, of the function $f(t) = 30e^{-7t}$.

33 Find the relative rate of change, $f'(t)/f(t)$, of the function $f(t) = 6t^2$.

34 Find the relative rate of change, $f'(t)/f(t)$, of the function $f(t) = 35t^{-4}$.

35 Find the relative rate of change of $f(t) = 5e^{1.5t}$ using the formula $\dfrac{d}{dt} \ln f(t)$.

36 Find the relative rate of change of $f(t) = 6.8e^{-0.5t}$ using the formula $\dfrac{d}{dt} \ln f(t)$.

37 Find the relative rate of change of $f(t) = 3t^2$ using the formula $\dfrac{d}{dt} \ln f(t)$.

38 Find the relative rate of change of $f(t) = 4.5t^{-4}$ using the formula $\dfrac{d}{dt} \ln f(t)$.

43 The distance traveled, D in feet, is a function of time, t, in seconds, with $D = f(t) = \sqrt{t^3 + 1}$. Find $f(10)$, $f'(10)$, and the relative rate of change f'/f at $t = 10$. Interpret your answers in terms of distance traveled.

52 Some economists suggest that an extra year of education increases a person's wages, on average, by about 14%. Assume you could make 10 dollars per hour with your current level of education and that inflation increases wages at a continuous rate of 3.5% per year.

(a) How much would you make per hour with four additional years of education?

(b) What is the difference between your wages in 20 years' time with and without the additional four years of education?

(c) Is the difference you found in part (b) increasing with time? If so, at what rate? (Assume the number of additional years of education stays fixed at four.)

53 Show that if the graphs of $f(t)$ and $h(t) = Ae^{kt}$ are tangent at $t = a$, then k is the relative rate of change of f at $t = a$.

Answers · Section 3.3

3.3.1 The Chain Rule for Powers

Ex 1 $y' = 7(4t^2+1)^{7-1}(8t+0)$

$\boxed{y' = 56t(4t^2+1)^6}$

Using z: $y = z^7,\ z = 4t^2+1,\ \dfrac{dy}{dt} = \dfrac{dy}{dz}\cdot\dfrac{dz}{dt} = 7z^6\cdot(8t) = 7(4t^2+1)^6(8t)$

1 $7(4x^2+1)^6(8x) = 56x(4x^2+1)^6$

2 $f'(x) = 99(x+1)^{98}$

3 $R' = 4(q^2+1)^3(2q) = 8q(q^2+1)^3$

4 $w' = 100(t^2+1)^{99}(2t) = 200t(t^2+1)^{99}$

5 $w' = 100(t^3+1)^{99}(3t^2) = 300t^2(t^3+1)^{99}$

6 $w' = 3(5r-6)^2(5) = 15(5r-6)^2$

7 $y = (s^3+1)^{1/2} \Rightarrow y' = \frac{1}{2}(s^3+1)^{-1/2}(3s^2) = \dfrac{3s^2}{2\sqrt{s^3+1}}$

9 $C' = 12\cdot 3(3q^2-5)^2(6q) = 216q(3q^2-5)^2$

22 $Q' = 100(0.5)(t^2+5)^{-0.5}(2t) = \dfrac{100t}{\sqrt{t^2+5}}$

24 $y' = 2(5+e^x)e^x$

25 $P' = 0.5(1+\ln x)^{-0.5}\cdot\dfrac{1}{x} = \dfrac{0.5}{x\sqrt{1+\ln x}}$

26 $\frac{1}{2}(e^x+1)^{-1/2}e^x = \dfrac{e^x}{2\sqrt{e^x+1}}$

27 $f'(x) = \frac{1}{2}(1-x^2)^{-1/2}(-2x) = \dfrac{-x}{\sqrt{1-x^2}}$

28 $f'(\theta) = -(e^\theta+e^{-\theta})^{-2}(e^\theta-e^{-\theta}) = -\dfrac{e^\theta-e^{-\theta}}{(e^\theta+e^{-\theta})^2}$

39 $y_1 = f(2) = (1)^3 = 1 \Rightarrow (2,1)\quad f'(x) = 3(x-1)^2 \Rightarrow m = f'(2) = 3$

$y-1 = 3(x-2) \Rightarrow \boxed{y = 3x-5}$

3.3.2 The Chain Rule for Exponentials and Logarithms

Ex 2 i. $y' = 5(3t^3-t)^4(9t^2-1) = \boxed{5(9t^2-1)(3t^3-t)^4}$

ii. $y' = \dfrac{1}{q^2+1}(2q+0) = \boxed{\dfrac{2q}{q^2+1}}$

Ex 3 i. $w' = 5\cdot\dfrac{1}{2t^2+3}\cdot(4t+0) = \boxed{\dfrac{20t}{2t^2+3}}$

ii. $y = \left(1+2e^{5t}\right)^{1/2} \Rightarrow y' = \frac{1}{2}\left(1+2e^{5t}\right)^{-1/2}\left(0+2\cdot 5e^{5t}\right) = \frac{1}{2}\left(10e^{5t}\right)\left(1+2e^{5t}\right)^{-1/2} = \boxed{\dfrac{5e^{5t}}{\sqrt{1+2e^{5t}}}}$

iii. $y = \left(5x^2+7\right)^{-1/3} \Rightarrow y' = -\frac{1}{3}\left(5x^2+7\right)^{-\frac{1}{3}-1}(10x+0) = -\frac{1}{3}(10x)\left(5x^2+7\right)^{-4/3} = \boxed{\dfrac{-10x}{3\left(5x^2+7\right)^{4/3}}}$

8 $y' = -6x+2(3)e^{3x} = -6x+6e^{3x}$

10 $f'(x) = 30e^{5x}+e^{-x^2}(-2x) = 30e^{5x}-2xe^{-x^2}$

11 $y' = 5e^{5t+1}(5) = 25e^{5t+1}$

12 $w' = e^{-3t^2}(-6t) = -6te^{-3t^2}$

13 $w' = e^{\sqrt{s}} \cdot \frac{1}{2}s^{-1/2} = \dfrac{e^{\sqrt{s}}}{2\sqrt{s}}$

14 $y' = \dfrac{5}{5t+1}$

15 $f'(x) = \dfrac{-1}{1-x} = \dfrac{1}{x-1}$

16 $f'(t) = \dfrac{2t}{t^2+1}$

17 $f'(x) = \dfrac{0-(-e^{-x})}{1-e^{-x}} = \dfrac{e^{-x}}{1-e^{-x}}$

18 $f'(x) = \dfrac{e^x}{e^x+1}$

19 $f'(t) = 5 \cdot \dfrac{5}{5t+1} = \dfrac{25}{5t+1}$

20 $g'(t) = \dfrac{4}{4t+9}$

21 $y' = \dfrac{3}{3t+2}$

23 $y' = 5 + \dfrac{1}{x+2}$

40 $R'(q) = \dfrac{2000q}{1+1000q^2} \Rightarrow R'(10) = \dfrac{20{,}000}{100{,}001} \approx 0.2$
When 10 goods are sold, selling one more increases revenue by about 0.2.

41 $v = \dfrac{ds}{dt} = 20 \cdot \frac{1}{2}e^{t/2} = \boxed{10e^{t/2}}$

42 (a) $\dfrac{dB}{dt} = P\ln\left(1+\dfrac{r}{100}\right)\left(1+\dfrac{r}{100}\right)^t$: how fast the balance is growing, in dollars per year.
(b) $\dfrac{dB}{dr} = Pt\left(1+\dfrac{r}{100}\right)^{t-1} \cdot \dfrac{1}{100} = \dfrac{Pt}{100}\left(1+\dfrac{r}{100}\right)^{t-1}$: about how much more money (in dollars) you would have after t years if the interest rate were 1% higher.

3.3.3 The Chain Rule with Values and Graphs

Ex 4 i. $h'(x) = f'(g(x)) \cdot g'(x)$. At $x = 1$: $h'(1) = f'(g(1)) \cdot g'(1) = f'(0) \cdot (1) = (0)(1) = \boxed{0}$
ii. $k'(x) = g'(f(x)) \cdot f'(x)$. At $x = 1$: $k'(1) = g'(f(1)) \cdot f'(1) = g'(1) \cdot (3) = (1)(3) = \boxed{3}$

Ex 5 To find derivatives, use the slopes of the tangent lines (the lines in the graph).
i. $h'(1) = f'(g(1)) \cdot g'(1)$. From the graph $g(1) = 7$, $g'(1) = \dfrac{7-6}{1-2} = -1$, and $f'(7) = 0$ (horizontal line).
$h'(1) = f'(7) \cdot g'(1) = (0)(-1) = \boxed{0}$
ii. $k'(2) = g'(f(2)) \cdot f'(2)$. From the graph $f(2) = 4$, $f'(2) = \dfrac{4-2}{2-1} = 2$, $g'(4) = \dfrac{3-4}{5-4} = -1$.
$k'(2) = g'(4) \cdot f'(2) = (-1)(2) = \boxed{-2}$

44 From the graphs: f has slope -2 on $0 < x < 40$ and slope 1 on $40 < x < 80$; g has slope $\frac{1}{2}$ on $0 < x < 40$ and slope 0 on $40 < x < 80$.
$\dfrac{d}{dx}f(g(x))\Big|_{x=30} = f'(g(30)) \cdot g'(30) = f'(55) \cdot \frac{1}{2} = (1)\left(\frac{1}{2}\right) = \boxed{\frac{1}{2}}$

45 $f'(g(70)) \cdot g'(70) = f'(60) \cdot 0 = \boxed{0}$

46 $g'(f(30)) \cdot f'(30) = g'(20) \cdot (-2) = \left(\frac{1}{2}\right)(-2) = \boxed{-1}$

47 $g'(f(70)) \cdot f'(70) = g'(30) \cdot (1) = \left(\frac{1}{2}\right)(1) = \boxed{\frac{1}{2}}$

48 $h'(1) = f'(g(1)) \cdot g'(1)$. From the graph $g(1) \approx 2$, $g'(1) \approx 0.8$, $f'(2) \approx 1$.
$h'(1) \approx (1)(0.8) \approx \boxed{0.8}$ (estimate)

49 $k'(1) = g'(f(1)) \cdot f'(1)$. From the graph $f(1) \approx -0.4$, $f'(1) \approx 0.5$, $g'(-0.4) \approx 2$.
$k'(1) \approx (2)(0.5) \approx \boxed{1}$

50 $h'(2) = f'(g(2)) \cdot g'(2)$. From the graph $g(2) \approx 1.6$, $g'(2) \approx -2$, $f'(1.6) \approx 0.7$.
$h'(2) \approx (0.7)(-2) \approx \boxed{-1.4}$ (estimate)

51 $k'(2) = g'(f(2)) \cdot f'(2)$. From the graph $f(2) \approx 0.3$, $f'(2) \approx 1$, $g'(0.3) \approx 1.9$.
$k'(2) \approx (1.9)(1) \approx \boxed{1.9}$

3.3.4 Relative Rates and Logarithms

Ex 6 $\text{RRC} = \dfrac{z'(t)}{z(t)} = \dfrac{P_0 k e^{kt}}{P_0 e^{kt}} \times 100\% = \boxed{k \times 100\%}$

Ex 7 $\text{RRC} = \dfrac{S'}{S} = \dfrac{1095\left(\frac{2}{3}\right) M^{\frac{2}{3}-1}}{1095 M^{2/3}} = \dfrac{\frac{2}{3} M^{-1/3}}{M^{2/3}} = \dfrac{2}{3M^{\frac{2}{3}+\frac{1}{3}}}$

$\boxed{\text{RRC} = \dfrac{2}{3M} \times 100\% \text{ per kg}}$

When $M = 70$ kg: $\text{RRC} = \dfrac{2}{3(70)} \times 100\% \approx 0.95\%$ per kg

The surface area of a human with body mass 70 kilograms increases by about 0.95% if body mass increases by 1 kilogram.

29 $\dfrac{f'}{f} = \dfrac{10}{10t + 5}$

30 $\dfrac{f'}{f} = \dfrac{15}{15t + 12}$

31 $\dfrac{40e^{5t}}{8e^{5t}} = 5$

32 $\dfrac{-210e^{-7t}}{30e^{-7t}} = -7$

33 $\dfrac{12t}{6t^2} = \dfrac{2}{t}$

34 $\dfrac{-140t^{-5}}{35t^{-4}} = -\dfrac{4}{t}$

35 $\ln f = \ln 5 + 1.5t \Rightarrow \dfrac{d}{dt} \ln f = 1.5$

36 $\ln f = \ln 6.8 - 0.5t \Rightarrow \dfrac{d}{dt} \ln f = -0.5$

37 $\ln f = \ln 3 + 2 \ln t \Rightarrow \dfrac{d}{dt} \ln f = \dfrac{2}{t}$

38 $\ln f = \ln 4.5 - 4 \ln t \Rightarrow \dfrac{d}{dt} \ln f = -\dfrac{4}{t}$

43 $f(10) = \sqrt{1001} \approx 31.64$ feet: after 10 seconds the body has traveled about 31.64 feet.
$f'(t) = \frac{1}{2}(t^3+1)^{-1/2}(3t^2) = \dfrac{3t^2}{2\sqrt{t^3+1}} \Rightarrow f'(10) = \dfrac{300}{2\sqrt{1001}} \approx 4.741$ ft/sec: at 10 seconds it is moving at about 4.74 feet per second.
$\dfrac{f'(10)}{f(10)} = \dfrac{4.741}{31.64} \approx 0.15 = 15\%$ per second: the distance is increasing at about 15% per second.

52 (a) $10(1.14)^4 \approx 16.89$ dollars per hour.
(b) With: $16.89e^{0.035(20)}$; without: $10e^{0.035(20)}$. Difference $= (16.89-10)e^{0.7} \approx 6.89(2.014) \approx 13.87$ dollars per hour.
(c) $D(t) = 6.89e^{0.035t}$ is increasing. $D'(t) = 0.035(6.89)e^{0.035t} \Rightarrow D'(20) \approx 0.49$ dollars per hour per year.

53 Tangent at $t = a$ means same value and same slope: $f(a) = Ae^{ka}$ and $f'(a) = h'(a) = kAe^{ka}$.
So $\dfrac{f'(a)}{f(a)} = \dfrac{kAe^{ka}}{Ae^{ka}} = k.$

3.4 The Product and Quotient Rules

In this section

1. The Product Rule 2 lecture · 25 textbook
2. The Quotient Rule 1 lecture · 9 textbook
3. Product and Quotient Rules with Values and Graphs 2 lecture · 5 textbook
4. Tangent Lines and Applications 0 lecture · 8 textbook

3.4.1 The Product Rule

Product rule	$(fg)' = f'g + fg'$ or $\dfrac{d(uv)}{dx} = \dfrac{du}{dx} \cdot v + u \cdot \dfrac{dv}{dx}$
In words	Derivative of the first times the second, plus the first times the derivative of the second.

Lecture Problems

Ex 1 Differentiate

i. $x^2 e^{2x}$

ii. $t^3 \ln(t+1)$

iii. $(3x^2 + 5x)e^x$

Ex 2 Find the derivative of $C = \dfrac{e^{2t}}{t}$.

Textbook Problems

Find the derivative. Assume that a, b, c, and k are constants.

3 $f(x) = xe^x$

4 $f(t) = te^{-2t}$

5 $y = 5xe^{x^2}$

6 $y = t^2(3t + 1)^3$

7 $y = x \ln x$

8 $y = (t^2 + 3)e^t$

9 $z = (3t + 1)(5t + 2)$

10 $y = (t^3 - 7t^2 + 1)e^t$

11 $P = t^2 \ln t$

12 $R = 3qe^{-q}$

15 $y = te^{-t^2}$

16 $f(z) = \sqrt{z}\, e^{-z}$

17 $g(p) = p \ln(2p + 1)$

18 $f(t) = te^{5-2t}$

19 $f(w) = (5w^2 + 3)e^{w^2}$

20 $y = x \cdot 2^x$

21 $w = (t^3 + 5t)(t^2 - 7t + 2)$

22 $z = (te^{3t} + e^{5t})^9$

30 $f(x) = (ax^2 + b)^3$

31 $f(x) = axe^{-bx}$

32 $f(t) = ae^{bt}$

33 $g(\alpha) = e^{\alpha e^{-2\alpha}}$

1 If $f(x) = (2x + 1)(3x - 2)$, find $f'(x)$ two ways: by using the product rule and by multiplying out. Do you get the same result?

2 If $f(x) = x^2(x^3 + 5)$, find $f'(x)$ two ways: by using the product rule and by multiplying out before taking the derivative. Do you get the same result? Should you?

34 If $f(x) = (3x + 8)(2x - 5)$, find $f'(x)$ and $f''(x)$.

3.4.2 The Quotient Rule

Quotient rule $\left(\frac{f}{g}\right)' = \frac{f'g - fg'}{g^2}$ or $\frac{d}{dx}\left(\frac{u}{v}\right) = \frac{\frac{du}{dx}\cdot v - u\cdot\frac{dv}{dx}}{v^2}$

Constant on top If $f(x) = \frac{c}{u(x)}$ (c constant), then $f'(x) = \frac{-c\,u'(x)}{\left(u(x)\right)^2}$

Lecture Problems

Ex 3 Differentiate

i. $\frac{5x^2}{x^3+1}$

ii. $\frac{1}{1+e^x}$

Textbook Problems

Find the derivative. Assume that a, b, c, and k are constants.

13 $f(t) = \frac{5}{t} + \frac{6}{t^2}$

14 $f(x) = \frac{x^2+3}{x}$

23 $f(x) = \frac{x}{e^x}$

24 $w = \frac{3z}{1+2z}$

25 $z = \frac{1-t}{1+t}$

26 $y = \frac{e^x}{1+e^x}$

27 $w = \frac{3y+y^2}{5+y}$

28 $y = \frac{1+z}{\ln z}$

29 $f(x) = \dfrac{ax+b}{cx+k}$

3.4.3 Product and Quotient Rules with Values and Graphs

Product rule $(fg)' = f'g + fg'$ or $\dfrac{d(uv)}{dx} = \dfrac{du}{dx}\cdot v + u\cdot\dfrac{dv}{dx}$

Quotient rule $\left(\dfrac{f}{g}\right)' = \dfrac{f'g - fg'}{g^2}$ or $\dfrac{d}{dx}\left(\dfrac{u}{v}\right) = \dfrac{\frac{du}{dx}\cdot v - u\cdot\frac{dv}{dx}}{v^2}$

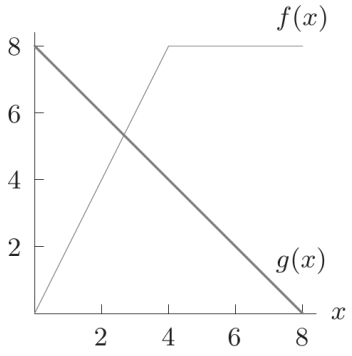
Lecture Problems

Ex 4 If $f(2) = 1$, $f'(2) = 5$, $g(2) = 3$ and $g'(2) = 6$, find

i. $h'(2)$ if $h(x) = f(x)g(x)$

ii. $k'(2)$ if $k(x) = f(x)/g(x)$

Ex 5 Let $h(x) = f(x)g(x)$ and $k(x) = f(x)/g(x)$. Use the given figure to estimate



Figure

i. $h'(2)$

ii. $k'(6)$

Textbook Problems

43 If $\dfrac{d}{dt}\big(tf(t)\big) = 1 + f(t)$, what is $f'(t)$?

44 The quantity, q, of a certain skateboard sold depends on the selling price, p, in dollars, so we write $q = f(p)$. You are given that $f(140) = 15{,}000$ and $f'(140) = -100$.

(a) What do $f(140) = 15{,}000$ and $f'(140) = -100$ tell you about the sales of skateboards?

(b) The total revenue, R, earned by the sale of skateboards is given by $R = pq$. Find $\left.\dfrac{dR}{dp}\right|_{p=140}$.

(c) What is the sign of $\left.\dfrac{dR}{dp}\right|_{p=140}$? If the skateboards are currently selling for 140 dollars, what happens to revenue if the price is increased to 141 dollars?

45 Show that the relative rate of change of a product fg is the sum of the relative rates of change of f and g.

46 Show that the relative rate of change of a quotient f/g is the difference between the relative rates of change of f and g.

47 If $h = f^n$, show that $\dfrac{(f^n)'}{f^n} = n\dfrac{f'}{f}$.

3.4.4 Tangent Lines and Applications

Textbook Problems

35 Find the equation of the tangent line to the graph of $f(x) = x^2 e^{-x}$ at $x = 0$. Check by graphing this function and the tangent line on the same axes.

36 Find the equation of the tangent line to the graph of $f(x) = \dfrac{2x - 5}{x + 1}$ at the point at which $x = 0$.

37 The quantity of a drug, Q mg, present in the body t hours after an injection of the drug is given is $Q = f(t) = 100te^{-0.5t}$. Find $f(1)$, $f'(1)$, $f(5)$, and $f'(5)$. Give units and interpret the answers.

38 A drug concentration curve is given by $C = f(t) = 20te^{-0.04t}$, with C in mg/ml and t in minutes.

(a) Graph C against t. Is $f'(15)$ positive or negative? Is $f'(45)$ positive or negative? Explain.

(b) Find $f(30)$ and $f'(30)$ analytically. Interpret them in terms of the concentration of the drug in the body.

39 For positive constants c and k, the *Monod growth curve* describes the growth of a population, P, as a function of the available quantity of a resource, r: $P = \dfrac{cr}{k + r}$. Find dP/dr and interpret it in terms of the growth of the population.

---

40 If p is price in dollars and q is quantity, demand for a product is given by $q = 5000e^{-0.08p}$.

(a) What quantity is sold at a price of 10 dollars?

(b) Find the derivative of demand with respect to price when the price is 10 dollars and interpret your answer in terms of demand.

---

41 The demand for a product is given in Problem 40. Find the revenue and the derivative of revenue with respect to price at a price of 10 dollars. Interpret your answers in economic terms.

---

42 The quantity demanded of a certain product, q, is given in terms of p, the price, by $q = 1000e^{-0.02p}$.

(a) Write revenue, R, as a function of price.

(b) Find the rate of change of revenue with respect to price.

(c) Find the revenue and rate of change of revenue with respect to price when the price is 10 dollars. Interpret your answers in economic terms.

Answers · Section 3.4

3.4.1 The Product Rule

Ex 1 i. $y' = (x^2)'(e^{2x}) + (x^2)(e^{2x})' = 2xe^{2x} + x^2(2e^{2x}) = 2xe^{2x} + 2x^2e^{2x} = \boxed{2xe^{2x}(1+x)}$

ii. $w' = (t^3)'\ln(t+1) + t^3\big(\ln(t+1)\big)' = 3t^2\ln(t+1) + t^3 \cdot \dfrac{1}{t+1} = \boxed{3t^2\ln(t+1) + \dfrac{t^3}{t+1}}$

iii. $\big((3x^2+5x)e^x\big)' = (3x^2+5x)'e^x + (3x^2+5x)(e^x)' = \boxed{(6x+5)e^x + (3x^2+5x)e^x}$

Ex 2 **Product rule:** $C = e^{2t}\cdot t^{-1}$

$$C' = (e^{2t})'t^{-1} + e^{2t}(t^{-1})' = 2e^{2t}t^{-1} + e^{2t}(-t^{-2}) = e^{2t}\left(\frac{2t}{t^2} - \frac{1}{t^2}\right) = \boxed{e^{2t}\left(\frac{2t-1}{t^2}\right)}$$

Quotient rule: $C' = \dfrac{(e^{2t})'t - e^{2t}(t)'}{t^2} = \dfrac{2e^{2t}\cdot t - e^{2t}\cdot 1}{t^2} = \boxed{\dfrac{e^{2t}(2t-1)}{t^2}}$

3 $f'(x) = e^x + xe^x = e^x(x+1)$

4 $f'(t) = e^{-2t} + t(-2e^{-2t}) = e^{-2t}(1-2t)$

5 $y' = 5e^{x^2} + 5x(2xe^{x^2}) = 5e^{x^2} + 10x^2e^{x^2}$

6 $y' = 2t(3t+1)^3 + t^2\cdot 3(3t+1)^2(3) = 2t(3t+1)^3 + 9t^2(3t+1)^2$

7 $y' = \ln x + x\cdot\dfrac{1}{x} = \ln x + 1$

8 $y' = 2te^t + (t^2+3)e^t = (t^2+2t+3)e^t$

9 $z' = 3(5t+2) + (3t+1)(5) = 30t + 11$

10 $y' = (3t^2-14t)e^t + (t^3-7t^2+1)e^t = (t^3-4t^2-14t+1)e^t$

11 $P' = 2t\ln t + t^2\cdot\dfrac{1}{t} = 2t\ln t + t$

12 $R' = 3e^{-q} + 3q(-e^{-q}) = 3e^{-q}(1-q)$

15 $y' = e^{-t^2} + t(-2te^{-t^2}) = e^{-t^2}(1-2t^2)$

16 $f'(z) = \frac{1}{2}z^{-1/2}e^{-z} - \sqrt{z}\,e^{-z} = e^{-z}\left(\dfrac{1}{2\sqrt{z}} - \sqrt{z}\right)$

17 $g'(p) = \ln(2p+1) + p\cdot\dfrac{2}{2p+1} = \ln(2p+1) + \dfrac{2p}{2p+1}$

18 $f'(t) = e^{5-2t} + t(-2e^{5-2t}) = e^{5-2t}(1-2t)$

19 $f'(w) = 10we^{w^2} + (5w^2+3)(2we^{w^2}) = 2we^{w^2}(5w^2+8)$

20 $y' = 2^x + x(\ln 2)2^x = 2^x(1 + x\ln 2)$

21 $w' = (3t^2+5)(t^2-7t+2) + (t^3+5t)(2t-7)$

22 $z' = 9(te^{3t}+e^{5t})^8\left(e^{3t}+3te^{3t}+5e^{5t}\right)$

30 $f'(x) = 3(ax^2+b)^2(2ax) = 6ax(ax^2+b)^2$

31 $f'(x) = ae^{-bx} + ax(-be^{-bx}) = ae^{-bx}(1-bx)$

32 $f'(t) = abe^{bt}$

33 $g'(\alpha) = e^{\alpha e^{-2\alpha}}\left(e^{-2\alpha} - 2\alpha e^{-2\alpha}\right) = (1-2\alpha)e^{-2\alpha}e^{\alpha e^{-2\alpha}}$

1 Product rule: $f'(x) = 2(3x-2) + (2x+1)(3) = 6x - 4 + 6x + 3 = 12x - 1$
Multiplying out: $f(x) = 6x^2 - x - 2 \Rightarrow f'(x) = 12x - 1$. Yes, the same result.

2 Product rule: $f'(x) = 2x(x^3+5) + x^2(3x^2) = 2x^4 + 10x + 3x^4 = 5x^4 + 10x$
Multiplying out: $f(x) = x^5 + 5x^2 \Rightarrow f'(x) = 5x^4 + 10x$. Yes, and we should, because it is the same function.

34 $f'(x) = 3(2x-5) + (3x+8)(2) = 12x + 1 \quad f''(x) = 12$

3.4.2 The Quotient Rule

Ex 3 i. $y' = \dfrac{(5x^2)'(x^3+1) - (5x^2)(x^3+1)'}{(x^3+1)^2} = \dfrac{10x(x^3+1) - 5x^2(3x^2)}{(x^3+1)^2} = \dfrac{10x^4 + 10x - 15x^4}{(x^3+1)^2} = \boxed{\dfrac{10x - 5x^4}{(x^3+1)^2}}$

ii. $w' = \dfrac{(1)'(1+e^x) - (1)(1+e^x)'}{(1+e^x)^2} = \dfrac{0(1+e^x) - e^x}{(1+e^x)^2} = \boxed{\dfrac{-e^x}{(1+e^x)^2}}$

13 $f(t) = 5t^{-1} + 6t^{-2} \Rightarrow f'(t) = -\dfrac{5}{t^2} - \dfrac{12}{t^3}$

14 $f'(x) = \dfrac{2x \cdot x - (x^2+3)(1)}{x^2} = \dfrac{x^2 - 3}{x^2}$

23 $f'(x) = \dfrac{e^x - xe^x}{e^{2x}} = \dfrac{1-x}{e^x}$

24 $w' = \dfrac{3(1+2z) - 3z(2)}{(1+2z)^2} = \dfrac{3}{(1+2z)^2}$

25 $z' = \dfrac{(-1)(1+t) - (1-t)(1)}{(1+t)^2} = \dfrac{-2}{(1+t)^2}$

26 $y' = \dfrac{e^x(1+e^x) - e^x \cdot e^x}{(1+e^x)^2} = \dfrac{e^x}{(1+e^x)^2}$

27 $w' = \dfrac{(3+2y)(5+y) - (3y+y^2)(1)}{(5+y)^2} = \dfrac{y^2 + 10y + 15}{(5+y)^2}$

28 $y' = \dfrac{(1)\ln z - (1+z)\frac{1}{z}}{(\ln z)^2} = \dfrac{\ln z - \frac{1}{z} - 1}{(\ln z)^2}$

29 $f'(x) = \dfrac{a(cx+k) - (ax+b)c}{(cx+k)^2} = \dfrac{ak - bc}{(cx+k)^2}$

3.4.3 Product and Quotient Rules with Values and Graphs

Ex 4 i. $h'(x) = f'(x)g(x) + f(x)g'(x)$. At $x = 2$: $h'(2) = f'(2)g(2) + f(2)g'(2) = (5)(3) + (1)(6) = 15 + 6 = \boxed{21}$

ii. $k'(x) = \dfrac{f'(x)g(x) - f(x)g'(x)}{\big(g(x)\big)^2}$. At $x = 2$: $k'(2) = \dfrac{(5)(3) - (1)(6)}{(3)^2} = \dfrac{15-6}{9} = \boxed{1}$

Ex 5 i. From the graph: $f(2) = 4$, $f'(2) = \dfrac{2-4}{1-2} = 2$; $\quad g(2) = 6$, $g'(2) = \dfrac{7-6}{1-2} = -1$.
$h'(2) = f'(2)g(2) + f(2)g'(2) = (2)(6) + (4)(-1) = \boxed{8}$

ii. From the graph: $f(6) = 8$, $f'(6) = 0$ (horizontal line); $\quad g(6) = 2$, $g'(6) = \dfrac{3-2}{5-6} = -1$.
$k'(6) = \dfrac{f'(6)g(6) - f(6)g'(6)}{\big(g(6)\big)^2} = \dfrac{(0)(2) - (8)(-1)}{(2)^2} = \dfrac{8}{4} = \boxed{2}$

43 Product rule: $\dfrac{d}{dt}\big(tf(t)\big) = f(t) + tf'(t)$.

$f(t) + tf'(t) = 1 + f(t) \Rightarrow tf'(t) = 1 \Rightarrow \boxed{f'(t) = \dfrac{1}{t}}$

44 (a) At a price of 140 dollars, 15,000 skateboards are sold; raising the price by 1 dollar decreases sales by about 100 skateboards.

(b) $R = p\,f(p) \Rightarrow \dfrac{dR}{dp} = f(p) + pf'(p) \Rightarrow \left.\dfrac{dR}{dp}\right|_{p=140} = 15{,}000 + 140(-100) = 1000$

(c) Positive. Raising the price to 141 dollars increases revenue by about 1000 dollars.

45 $\dfrac{(fg)'}{fg} = \dfrac{f'g + fg'}{fg} = \dfrac{f'g}{fg} + \dfrac{fg'}{fg} = \boxed{\dfrac{f'}{f} + \dfrac{g'}{g}}$

46 $\dfrac{(f/g)'}{f/g} = \dfrac{f'g - fg'}{g^2} \cdot \dfrac{g}{f} = \dfrac{f'g - fg'}{fg} = \boxed{\dfrac{f'}{f} - \dfrac{g'}{g}}$

47 Chain rule: $(f^n)' = nf^{n-1}f'$, so $\dfrac{(f^n)'}{f^n} = \dfrac{nf^{n-1}f'}{f^n} = \boxed{n\dfrac{f'}{f}}$

3.4.4 Tangent Lines and Applications

35 $y_1 = f(0) = 0 \Rightarrow (0,0)\quad f'(x) = 2xe^{-x} - x^2e^{-x} \Rightarrow m = f'(0) = 0$

$y - 0 = 0(x - 0) \Rightarrow \boxed{y = 0}$ (the x-axis; the graph touches it at its minimum at $x = 0$).

36 $y_1 = f(0) = -5 \Rightarrow (0, -5)$

$f'(x) = \dfrac{2(x+1) - (2x-5)(1)}{(x+1)^2} = \dfrac{7}{(x+1)^2} \Rightarrow m = f'(0) = 7$

$y - (-5) = 7(x - 0) \Rightarrow \boxed{y = 7x - 5}$

37 $f'(t) = 100e^{-0.5t} + 100t(-0.5e^{-0.5t}) = 100e^{-0.5t}(1 - 0.5t)$

$f(1) = 100e^{-0.5} \approx 60.65$ mg: 1 hour after the injection there are about 60.65 mg in the body.

$f'(1) = 50e^{-0.5} \approx 30.33$ mg/hr: at 1 hour the quantity is increasing by about 30.33 mg per hour.

$f(5) = 500e^{-2.5} \approx 41.04$ mg; $\quad f'(5) = 100e^{-2.5}(-1.5) \approx -12.31$ mg/hr: at 5 hours there are about 41.04 mg and it is decreasing by about 12.31 mg per hour.

38 $f'(t) = 20e^{-0.04t} + 20t(-0.04e^{-0.04t}) = 20e^{-0.04t}(1 - 0.04t)$, which is 0 at $t = 25$.

(a) The graph rises to a maximum at $t = 25$ and then falls. So $f'(15) > 0$ (increasing) and $f'(45) < 0$ (decreasing).

(b) $f(30) = 600e^{-1.2} \approx 180.7$ mg/ml: after 30 minutes the concentration is about 180.7 mg/ml.

$f'(30) = 20e^{-1.2}(1 - 1.2) \approx -1.20$ (mg/ml)/min: at 30 minutes the concentration is decreasing by about 1.2 mg/ml per minute.

39 $\dfrac{dP}{dr} = \dfrac{c(k+r) - cr(1)}{(k+r)^2} = \boxed{\dfrac{ck}{(k+r)^2}}$

This is the approximate change in the population per unit increase in the resource. It is always positive (more resource, larger population) and gets smaller as r grows.

40 (a) $q = 5000e^{-0.8} \approx 2247$ items.

(b) $\dfrac{dq}{dp} = 5000(-0.08)e^{-0.08p} = -400e^{-0.08p} \Rightarrow \left.\dfrac{dq}{dp}\right|_{p=10} = -400e^{-0.8} \approx -180$ items per dollar: raising the price from 10 to 11 dollars reduces sales by about 180 items.

41 $R = pq = 5000pe^{-0.08p}$ $\quad R(10) = 50{,}000e^{-0.8} \approx 22{,}466$ dollars.

$R'(p) = 5000e^{-0.08p} + 5000p(-0.08e^{-0.08p}) = 5000e^{-0.08p}(1 - 0.08p)$

$R'(10) = 5000e^{-0.8}(0.2) \approx 449$ dollars per dollar: at a price of 10 dollars the revenue is about 22,466 dollars, and raising the price by 1 dollar increases revenue by about 449 dollars.

42 (a) $R = pq = 1000pe^{-0.02p}$

(b) $R'(p) = 1000e^{-0.02p} + 1000p(-0.02e^{-0.02p}) = 1000e^{-0.02p}(1 - 0.02p)$

(c) $R(10) = 10{,}000e^{-0.2} \approx 8187$ dollars; $\quad R'(10) = 1000e^{-0.2}(0.8) \approx 655$ dollars per dollar. At 10 dollars the revenue is about 8187 dollars, and a 1 dollar price increase raises revenue by about 655 dollars.

3.5 Derivatives of Periodic Functions

In this section

1. Derivatives of Sine and Cosine 2 lecture · 8 textbook
2. The Chain Rule with Sine and Cosine 1 lecture · 15 textbook
3. Applications of Periodic Functions 1 lecture · 7 textbook

3.5.1 Derivatives of Sine and Cosine

Sine and cosine For x in radians: $\dfrac{d}{dx}(\sin x) = \cos x$ and $\dfrac{d}{dx}(\cos x) = -\sin x$

Lecture Problems

Ex 1 Using a calculator, estimate the derivative of $y = \sin x$ at $x = 0$. Make sure your calculator is set in radians.

Ex 2 Differentiate

i. $5\sin t - 8\cos t$

ii. $5 - 3\sin x + x^3$

Textbook Problems

Differentiate. Assume that A and B are constants.

1 $y = 5\sin x$

2 $P = 3 + \cos t$

3 $y = t^2 + 5\cos t$

4 $y = B + A\sin t$

5 $R(q) = q^2 - 2\cos q$

6 $y = 5\sin x - 5x + 4$

21 Find the equation of the tangent line to the graph of $y = \sin x$ at $x = \pi$. Graph the function and the tangent line on the same axes.

24 Find equations of the tangent lines to the graph of $f(x) = \sin x$ at $x = 0$ and at $x = \pi/3$. Use each tangent line to approximate $\sin(\pi/6)$. Would you expect these results to be equally accurate, since they are taken equally far away from $x = \pi/6$ but on opposite sides? If the accuracy is different, can you account for the difference?

3.5.2 The Chain Rule with Sine and Cosine

Chain rule	$\frac{d}{dt}\big(\sin f(t)\big) = f'(t)\cos f(t)$ and $\frac{d}{dt}\big(\cos f(t)\big) = -f'(t)\sin f(t)$
Constant k	$\frac{d}{dt}(\sin kt) = k\cos kt$ and $\frac{d}{dt}(\cos kt) = -k\sin kt$

Lecture Problems

Ex 3 Differentiate

i. $\sin(t^2)$

ii. $5\cos(2t)$

iii. $2x\sin(3x)$

Textbook Problems

Differentiate. Assume that A and B are constants.

7 $f(x) = \sin(3x)$

8 $R = \sin(5t)$

9 $W = 4\cos(t^2)$

10 $y = 2\cos(5t)$

11 $y = \sin(x^2)$

12 $y = A\sin(Bt)$

13 $z = \cos(4\theta)$

14 $y = 6\sin(2t) + \cos(4t)$

15 $f(x) = x^2\cos x$

16 $f(x) = 2x\sin(3x)$

17 $f(\theta) = \theta^3 \cos\theta$

18 $z = \dfrac{e^{t^2} + t}{\sin(2t)}$

19 $f(t) = \dfrac{t^2}{\cos t}$

20 $f(\theta) = \dfrac{\sin\theta}{\theta}$

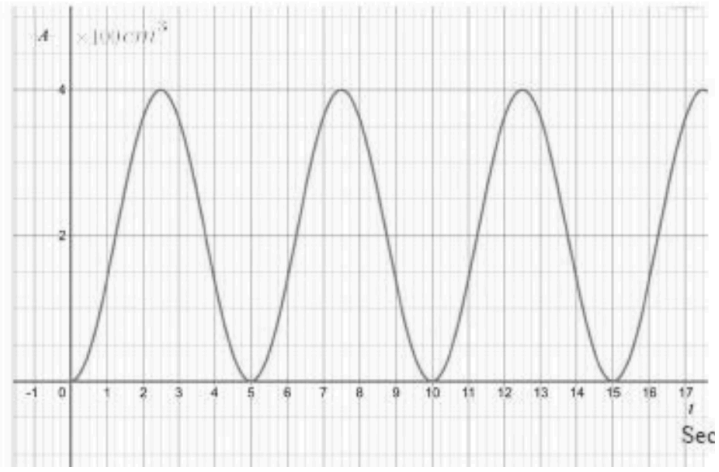
23 Is the graph of $y = \sin(x^4)$ increasing or decreasing when $x = 10$? Is it concave up or concave down?

3.5.3 Applications of Periodic Functions

Period For $\cos(Bt)$ or $\sin(Bt)$: period $P = \dfrac{2\pi}{B}$

Lecture Problems

Ex 4 The average adult takes about 12 breaths per minute. As a patient inhales, the volume of air in the lung increases. As the patient exhales, the volume of air in the lung decreases. For t in seconds since start of the breathing cycle, the volume of air inhaled or exhaled since $t = 0$ is given, in hundreds of cubic centimeters, by $A(t) = -2\cos\left(\dfrac{2\pi}{5}t\right) + 2$.



Volume of air, A (hundreds of cm³), against time t (seconds)

i. How long is one breathing cycle?

ii. Find $A'(1)$ and explain what it means.

Textbook Problems

22 If t is the number of months since June, the number of bird species, N, found in an Ohio forest oscillates approximately according to the formula $N = f(t) = 19 + 9\cos\left(\frac{\pi}{6}t\right)$.

(a) Graph $f(t)$ for $0 \le t \le 24$ and describe what it shows. Use the graph to decide whether $f'(1)$ and $f'(10)$ are positive or negative.

(b) Find $f'(t)$.

(c) Find and interpret $f(1)$, $f'(1)$, $f(10)$, and $f'(10)$.

25 A company's monthly sales, $S(t)$, are seasonal and given as a function of time, t, in months, by $S(t) = 2000 + 600\sin\left(\frac{\pi}{6}t\right)$.

(a) Graph $S(t)$ for $t = 0$ to $t = 12$. What is the maximum monthly sales? What is the minimum monthly sales? If $t = 0$ is January 1, when during the year are sales highest?

(b) Find $S(2)$ and $S'(2)$. Interpret in terms of sales.

26 A boat at anchor is bobbing up and down in the sea. The vertical distance, y, in feet, between the sea floor and the boat is given as a function of time, t, in minutes, by $y = 15 + \sin(2\pi t)$.

(a) Find the vertical velocity, v, of the boat at time t.

(b) Make rough sketches of y and v against t.

27 The depth of the water, y, in meters, in the Bay of Fundy, Canada, is given as a function of time, t, in hours after midnight, by the function $y = 10 + 7.5\cos(0.507t)$. How quickly is the tide rising or falling (in meters/hour) at each of the following times?

(a) 6:00 am

(b) 9:00 am

(c) Noon

(d) 6:00 pm

28 The average adult takes about 12 breaths per minute. As a patient inhales, the volume of air in the lung increases. As the patient exhales, the volume of air in the lung decreases. For t in seconds since start of the breathing cycle, the volume of air inhaled or exhaled since $t = 0$ is given, in hundreds of cubic centimeters, by $A(t) = -2\cos\left(\frac{2\pi}{5}t\right) + 2$.

(a) How long is one breathing cycle?

(b) Find $A'(1)$ and explain what it means.

29 On July 7, 2009, there was a full moon in the Eastern time zone. If t is the number of days since July 7th, the percent of moon illuminated can be represented by $H(t) = 50 + 50\cos\left(\frac{\pi}{15}t\right)$.

(a) Find $H'(t)$. Explain what this tells us about the moon.

(b) For $0 \leq t \leq 30$, when is $H'(t) = 0$? What does this tell us about the moon?

(c) For $0 \leq t \leq 30$, when is $H'(t)$ negative? When is it positive? Explain what positive and negative values of $H'(t)$ tell us about the moon.

30 Paris, France, has a latitude of approximately $49°$ N. If t is the number of days since the start of 2009, the number of hours of daylight in Paris can be approximated by $D(t) = 4\cos\left(\frac{2\pi}{365}(t - 172)\right) + 12$.

(a) Find $D(40)$ and $D'(40)$. Explain what this tells about daylight in Paris.

(b) Find $D(172)$ and $D'(172)$. Explain what this tells about daylight in Paris.

Answers · Section 3.5

3.5.1 Derivatives of Sine and Cosine

Ex 1 Use the average rate of change on the small interval $0 \le x \le 0.01$:

$$f'(0) \approx \frac{\sin(0.01) - \sin(0)}{0.01 - 0} = \frac{0.0099998 - 0}{0.01} = 0.99998 \approx 1.0$$

The derivative of $y = \sin x$ at $x = 0$ is approximately 1.0. *Warning:* x must be in radians.

Ex 2 i. $y' = 5\cos t - 8(-\sin t) = \boxed{5\cos t + 8\sin t}$

ii. $w' = 0 - 3\cos x + 3x^2 = \boxed{-3\cos x + 3x^2}$

1 $y' = 5\cos x$

2 $P' = -\sin t$

3 $y' = 2t - 5\sin t$

4 $y' = A\cos t$

5 $R'(q) = 2q + 2\sin q$

6 $y' = 5\cos x - 5$

21 $y_1 = \sin\pi = 0 \Rightarrow (\pi, 0)$ $\quad y' = \cos x \Rightarrow m = \cos\pi = -1$

$y - 0 = -1(x - \pi) \Rightarrow \boxed{y = -x + \pi}$

24 At $x = 0$: $f(0) = 0$, $m = \cos 0 = 1 \Rightarrow y = x$. Estimate: $\sin(\pi/6) \approx \pi/6 \approx 0.524$.

At $x = \pi/3$: $f = \frac{\sqrt{3}}{2}$, $m = \cos\frac{\pi}{3} = \frac{1}{2} \Rightarrow y = \frac{\sqrt{3}}{2} + \frac{1}{2}\left(x - \frac{\pi}{3}\right)$. Estimate: ≈ 0.604.

True value $\sin(\pi/6) = 0.5$. The estimate from $x = 0$ is more accurate, because near $x = 0$ the sine graph is almost straight (it bends very little), while near $x = \pi/3$ it bends more.

3.5.2 The Chain Rule with Sine and Cosine

Ex 3 i. $y' = \cos(t^2) \cdot (2t) = \boxed{2t\cos(t^2)}$

ii. $w' = 5\big(-\sin(2t) \cdot (2)\big) = \boxed{-10\sin(2t)}$

iii. $f' = (2x)'\sin(3x) + 2x\big(\sin(3x)\big)' = 2\sin(3x) + 2x\big(\cos(3x) \cdot 3\big) =$

$\boxed{2\sin(3x) + 6x\cos(3x)}$

7 $f'(x) = 3\cos(3x)$

8 $R' = 5\cos(5t)$

9 $W' = 4\big(-\sin(t^2)\big)(2t) = -8t\sin(t^2)$

10 $y' = -10\sin(5t)$

11 $y' = 2x\cos(x^2)$

12 $y' = AB\cos(Bt)$

13 $z' = -4\sin(4\theta)$

14 $y' = 12\cos(2t) - 4\sin(4t)$

15 $f'(x) = 2x\cos x - x^2\sin x$

16 $f'(x) = 2\sin(3x) + 6x\cos(3x)$

17 $f'(\theta) = 3\theta^2\cos\theta - \theta^3\sin\theta$

18 $z' = \dfrac{\left(2te^{t^2} + 1\right)\sin(2t) - 2\left(e^{t^2} + t\right)\cos(2t)}{\sin^2(2t)}$

19 $f'(t) = \dfrac{2t\cos t + t^2\sin t}{\cos^2 t}$

20 $f'(\theta) = \dfrac{\theta\cos\theta - \sin\theta}{\theta^2}$

23 $y' = 4x^3\cos(x^4) \Rightarrow y'(10) = 4000\cos(10{,}000) \approx -3809 < 0$: **decreasing**.
$y'' = 12x^2\cos(x^4) - 16x^6\sin(x^4) \Rightarrow y''(10) \approx 4.9 \times 10^6 > 0$: **concave up**.

3.5.3 Applications of Periodic Functions

Ex 4 i. $B = \dfrac{2\pi}{5}$, $P = \dfrac{2\pi}{B} = \boxed{5 \text{ seconds}}$

ii. $A'(t) = -2\left(-\sin\left(\dfrac{2\pi}{5}t\right)\cdot\dfrac{2\pi}{5}\right) + 0 = \dfrac{4\pi}{5}\sin\left(\dfrac{2\pi}{5}t\right)$

$A'(1) = \dfrac{4\pi}{5}\sin\left(\dfrac{2\pi}{5}\right) \approx 2.39$ hundred cm^3/sec

One second after the cycle begins, the patient is inhaling at a rate of about 2.39 hundred cubic centimeters per second, that is 239 cm^3/sec.

22 (a) The number of species oscillates between 10 and 28 with a period of 12 months (highest in June, lowest in December). At $t = 1$ the graph is decreasing, so $f'(1) < 0$; at $t = 10$ it is increasing, so $f'(10) > 0$.

(b) $f'(t) = -9\cdot\dfrac{\pi}{6}\sin\left(\dfrac{\pi}{6}t\right) = -\dfrac{3\pi}{2}\sin\left(\dfrac{\pi}{6}t\right)$

(c) $f(1) \approx 26.8$ species in July; $f'(1) \approx -2.36$ species per month (decreasing). $f(10) = 23.5$ species in April; $f'(10) \approx 4.08$ species per month (increasing).

25 (a) Maximum $2000 + 600 = 2600$; minimum $2000 - 600 = 1400$. Highest when $\sin\left(\frac{\pi}{6}t\right) = 1 \Rightarrow t = 3$: April 1.

(b) $S(2) = 2000 + 600\sin\left(\frac{\pi}{3}\right) \approx 2519.6$: on March 1 monthly sales are about 2520.

$S'(t) = 100\pi\cos\left(\frac{\pi}{6}t\right) \Rightarrow S'(2) = 50\pi \approx 157.1$ per month: sales are increasing by about 157 per month.

26 (a) $v = \dfrac{dy}{dt} = 2\pi\cos(2\pi t)$ feet per minute.

(b) y oscillates between 14 and 16 with period 1 minute; v oscillates between -2π and 2π with the same period, and $v = 0$ at the top and bottom of y.

27 $y' = -7.5(0.507)\sin(0.507t) = -3.8025\sin(0.507t)$

(a) $t = 6$: $y' \approx -0.38$: falling at 0.38 m/hr (b) $t = 9$: $y' \approx 3.76$: rising at 3.76 m/hr

(c) $t = 12$: $y' \approx 0.75$: rising at 0.75 m/hr (d) $t = 18$: $y' \approx -1.12$: falling at 1.12 m/hr

28 Same as lecture Ex 4: (a) 5 seconds (b) $A'(1) \approx 2.39$ hundred cm^3/sec: the patient is inhaling at about 239 cm^3 per second.

29 (a) $H'(t) = -\dfrac{10\pi}{3}\sin\left(\dfrac{\pi}{15}t\right)$: the rate (percent per day) at which the illuminated part of the moon is changing.

(b) $H'(t) = 0$ at $t = 0$, 15, 30 days: full moon ($t = 0$, 30) and new moon ($t = 15$); the percent illuminated is not changing at these moments.

(c) $H'(t) < 0$ for $0 < t < 15$: the illuminated part is shrinking. $H'(t) > 0$ for $15 < t < 30$: it is growing.

30 $D'(t) = -\dfrac{8\pi}{365}\sin\left(\dfrac{2\pi}{365}(t-172)\right)$

(a) $D(40) \approx 9.42$ hours; $D'(40) \approx 0.053$ hours per day: on day 40 (about February 10) there are about 9.4 hours of daylight, increasing by about 3 minutes per day.

(b) $D(172) = 16$ hours; $D'(172) = 0$: on day 172 (about June 21–22) there are 16 hours of daylight (the longest day), and the amount is momentarily not changing.